

BIOCHEMISTRY

Stoichiometry and architecture of the human pyruvate dehydrogenase complex

Rafal Zdanowicz¹, Pavel Afanasyev², Adam Pruška³, Julian A. Harrison³, Christoph Giese¹, Daniel Boehringer², Alexander Leitner⁴, Renato Zenobi³, Rudi Glockshuber^{1*}

The pyruvate dehydrogenase complex (PDHc) is a key megaenzyme linking glycolysis with the citric acid cycle. In mammalian PDHc, dihydrolipoamide acetyltransferase (E2) and the dihydrolipoamide dehydrogenase-binding protein (E3BP) form a 60-subunit core that associates with the peripheral subunits pyruvate dehydrogenase (E1) and dihydrolipoamide dehydrogenase (E3). The structure and stoichiometry of the fully assembled, mammalian PDHc or its core remained elusive. Here, we demonstrate that the human PDHc core is formed by 48 E2 copies that bind 48 E1 heterotetramers and 12 E3BP copies that bind 12 E3 homodimers. Cryo-electron microscopy, together with native and cross-linking mass spectrometry, confirmed a core model in which 8 E2 homotrimers and 12 E2-E3BP heterotrimers assemble into a pseudoicosahedral particle such that the 12 E3BP molecules form six E3BP-E3BP intertrimer interfaces distributed tetrahedrally within the 60-subunit core. The even distribution of E3 subunits in the peripheral shell of PDHc guarantees maximum enzymatic activity of the megaenzyme.

INTRODUCTION

α -Keto acid dehydrogenase complexes form a ubiquitous family of enzymes catalyzing oxidative decarboxylation of α -keto acids in glycolysis, the citric acid cycle, and branched-chain amino acid metabolism (1, 2). The pyruvate dehydrogenase complex (PDHc) is a multienzyme assembly that links the two major metabolic pathways of glycolysis and citric acid cycle, thereby mediating energy production by oxidative phosphorylation. PDHc converts the product of glycolysis, pyruvate, into acetyl-coenzyme A (acetyl-CoA) that enters the citric acid cycle, which is the source of reduced nicotinamide adenine dinucleotide (NADH) and flavin adenine dinucleotide (FADH₂) in cells (3–5). As the reaction catalyzed by PDHc is indispensable for mammalian cells, familial deficiencies in the activity of this complex are highly debilitating. Deficient PDHc variants may lead to metabolic acidosis, neurodevelopmental delay, hypotonia, and brain abnormalities, symptoms often associated with disorders like Leigh syndrome or ataxia (6–8). Moreover, PDHc and related α -keto acid dehydrogenase complexes were identified as sources of superoxide/H₂O₂ in mitochondria that may generate oxidative stress (9, 10).

PDHcs in prokaryotes and eukaryotes are composed of multiple copies of the enzymes pyruvate dehydrogenase (also termed E1 component), dihydrolipoamide acetyltransferase (E2 component), and dihydrolipoamide dehydrogenase (E3 component), which act sequentially to generate acetyl-CoA, NADH, H⁺, and CO₂ from pyruvate, NAD⁺, and CoA-SH. The thiamine diphosphate (ThDP)-dependent enzyme E1 catalyzes the decarboxylation of pyruvate to hydroxyethyl-ThDP, which reacts with the E2-bound cofactor lipoamide to S-acetyl-lipoamide. E2 then catalyzes the acetyl group transfer to CoA-SH, generating acetyl-CoA and dihydrolipoamide. Last, E3 regenerates E2-bound lipoamide by dihydrolipoamide

oxidation with NAD⁺, a reaction mediated by the E3-bound cofactor flavin adenine dinucleotide (FAD) (11–13).

The exact subunit composition of the fully assembled PDHc varies between species, but the common architectural principle of the complex, in which an oligomeric E2 core associates with multiple copies of E1 and E3 that form a peripheral shell around the core, is conserved (14). Within the complex, effective shuttling of catalytic intermediates between the active sites of E1, E2, and E3 is accomplished via long, flexible linkers. These unstructured linkers connect the N-terminal lipoyl domains and the peripheral subunit binding domain of E2 with its C-terminal catalytic domain, which is the only domain that contributes to the assembly of the E2 core (2, 5, 15, 16). While the individual reactions of the catalytic cycle are identical in prokaryotic and eukaryotic PDHcs, the complexes differ in molecular architecture and subunit composition. In Gammaproteobacteria (e.g., *Escherichia coli*), the PDHc core is a cubic ~1.5-MDa homooligomer of 24 E2 subunits (8 E2 homotrimers) to which a maximum of 16 E1 homodimers and 8 E3 homodimers can bind via the peripheral subunit binding domains of E2 under saturating conditions in vitro (17–19). In contrast, the mammalian PDHc core is composed of 60 subunits and exhibits a pseudoicosahedral symmetry with a mass of ~3.5 MDa (20). Moreover, the mammalian PDHc core is a hetero-oligomer containing not only E2 (exclusively binding E1 via its central E1-binding domain) but also the catalytically inactive, E3-binding protein E3BP (exclusively binding E3 via an analogous, central E3-binding domain) (21, 22). The assembly of the 60-subunit mammalian PDHc core occurs via interactions between the C-terminal, catalytic domains of E2 and the homologous (pseudocatalytic) C-terminal domains of E3BP (fig. S1, A and B) (23). In addition, while E1 from Gram-negative bacteria is a homodimer, mammalian E1 is an α_2/β_2 heterotetramer. Consequently, mammalian PDHcs are composed of five different polypeptide chains (E1 α , E1 β , E2, E3BP, and E3). Moreover, the rate-limiting, catalytic E1 reaction of mammalian PDHc is regulated by phosphorylation/dephosphorylation reactions catalyzed by pyruvate dehydrogenase kinases and pyruvate dehydrogenase phosphatases (24–27). Overall, the fully assembled mammalian PDHcs were reported to have a mass of ~9.5 MDa (5, 28).

¹Department of Biology, Institute of Molecular Biology and Biophysics, ETH Zurich, Otto-Stern-Weg 5, 8093 Zürich, Switzerland. ²Cryo-EM Knowledge Hub, ETH Zurich, Otto-Stern-Weg 3, 8093 Zürich, Switzerland. ³Department of Chemistry and Applied Biosciences, Laboratory of Organic Chemistry, ETH Zurich, Vladimir-Prelog-Weg 3, 8093 Zürich, Switzerland. ⁴Department of Biology, Institute of Molecular Systems Biology, ETH Zurich, Otto-Stern-Weg 3, 8093 Zürich, Switzerland.

*Corresponding author. Email: rudi@mol.biol.ethz.ch

Structural studies on mammalian PDHc proved to be particularly challenging since the pseudoicosahedral complex exhibits considerable intrinsic flexibility to allow for channeling catalytic intermediates between active sites (2, 16). Although structures of the isolated mammalian E1 heterotetramer (29), a nonphysiological E2 60-mer (23), and the E3 dimer (28) have been solved, no three-dimensional (3D) structure at atomic resolution is available for the fully assembled complex or the functional, hetero-oligomeric core composed of E2 and E3BP. There are also contradictory results on the E2:E3BP stoichiometry of the hetero-oligomeric core of human PDHc (termed hCore in the following). Specifically, two alternative “substitution” models have been proposed: the 40:20 (E2₄₀:E3BP₂₀) (30, 31) and 48:12 (E2₄₈:E3BP₁₂) (32, 33) models in which either 20 or 12 hE3BPs replace the equivalent number of hE2s within the hCore 60-mer, respectively.

Here, we focused on the biochemical and structural characterization of human PDHc with emphasis on the architecture of hCore and its binding to the peripheral subunits hE1 and hE3. Using size exclusion chromatography (SEC), fluorescence spectroscopy, gel densitometry, enzyme kinetics, and mass spectrometry (MS), we characterized the stoichiometry of hCore and its binding to hE1 and hE3. In addition, the closely related PDHc from the porcine liver (pPDHc) was purified and compared with recombinantly produced hCore. Last, the structural organization of hCore was analyzed by negative staining (NS-EM) and cryo-electron microscopy (cryo-EM), as well as native and cross-linking MS (XL-MS). The results show that the pseudoicosahedral human PDHc core exhibits an hE2:hE3BP stoichiometry of 48:12, with a tetrahedral arrangement of hE3BPs within the hCore particle. In addition, we demonstrate that the hPDHc particle is composed of hE2₃ homotrimers and heterotrimers formed by two hE2 subunits and one copy of the structurally homologous hE3BP subunit.

RESULTS

Recombinant production and purification of the human E2/E3BP (hCore) complex

We recombinantly coexpressed human E2 and human E3BP in *E. coli* and purified the in vivo assembled hCore complex via ammonium sulfate precipitation, sucrose gradient ultracentrifugation, and SEC. The established purification protocol proved to be highly effective, as the first step of salting out with 25% saturated ammonium sulfate already resulted in a ~90% pure sample. Sucrose gradient ultracentrifugation was used to further separate the 60-mer from larger aggregates and low molecular mass impurities. SEC, as the final purification step, was used to exchange the buffer, with hCore eluting in the void volume of the column (Fig. 1A). The purity and integrity of the hCore particles were verified by SDS-polyacrylamide gel electrophoresis (PAGE; inset in Fig. 1A) and NS-EM (Fig. 1B), and the completeness of lipoylation of both hE2 (two lipoyl domains) and hE3BP (one lipoyl domain) was confirmed by electrospray ionization MS (34, 35) (lipoylated hE2: calculated: 61,746.5 Da, measured: 61,748.5 Da; lipoylated hE3BP: 48,212.5 Da, measured: 48,212.5 Da) (Fig. 1C).

Composition of the porcine and human PDHc cores

To better understand the composition of hCore and to have a native reference for the recombinantly produced hCore particle, we purified PDHc from porcine liver (*Sus scrofa domestica*). The porcine

complexes isolated via multiple rounds of centrifugation, followed by sucrose gradient ultracentrifugation, and polyethylene glycol (PEG) precipitation (fig. S2) were examined by gel densitometry (Fig. 2A) to extract information on the relative amounts of E2 and E3BP. For this purpose, a calibration curve was generated using separately purified hE2 homo-60-mer and hE3BP mixed at different molar ratios. The equivalent approach was used to analyze the composition of the recombinantly produced hPDHc core. Although the porcine and human orthologs show 93 and 91% identity for E2 and E3BP, respectively, their sequences differ slightly in the number of basic amino acids to which the Coomassie blue stain binds. For that reason, besides using standard Coomassie staining, we also used stain-free gels (Bio-Rad) containing a trihalo compound that specifically cross-links with tryptophan residues upon ultraviolet (UV) irradiation, generating a strong fluorescence signal and offering a second, complementary detection method. The porcine and human PDHc core proteins have identical numbers of tryptophans, thus minimizing staining intensity errors. The averaged E2:E3BP band intensity ratios determined for the Coomassie-stained gels and stain-free detection are listed in Fig. 2B. Values obtained for both detection techniques and the two complexes were essentially identical and yielded 13 E3BP molecules in a 60-subunit assembly. Within the experimental error of protein concentration determination ($\leq 10\%$), our densitometry analysis of the native porcine and recombinantly produced human PDHc core is thus fully consistent with the 48:12 (E2:E3BP) model of the mammalian PDHc core.

Binding stoichiometry between the core and the peripheral subunits

To determine the binding stoichiometry between the recombinantly produced hCore hetero-oligomers and the peripheral hPDHc subunits, we performed a series of SEC runs after titrating hCore complexes with increasing amounts of hE1 or hE3 (Fig. 3, A and B). The binding stoichiometry was determined to be 50.1 ± 0.7 and 11.3 ± 1.1 molar equivalents (meq) of hE1 (heterotetramers) and hE3 (homodimers) to hCore, respectively. These data thus provide strong additional support for the 48:12 model of the core complex, indicating that all hE2 and hE3BP proteins within hCore can be fully saturated with hE1 and hE3 subunits, respectively. In addition, sedimentation velocity and equilibrium analytical ultracentrifugation showed that the main fraction of purified, recombinant hE3BP is a monomer, with no evidence for specific homo-oligomerization (fig. S3). As expected, purified hE3BP monomers bound hE3 dimers with a 1:1 stoichiometry (fig. S4A) but did not bind to or exchange with subunits of hE2 homo-60-mer (fig. S4B).

Affinities and kinetics of hE1/hE3 binding to the core complex

The affinities between hCore and the peripheral subunits were studied at 40-fold lower hCore concentration (2 nM) compared to the SEC analysis described above. The enzymatic activity of hPDHc was measured as a function of the meq of hE1 and hE3 added, using hCore saturated with the respective other peripheral subunit, i.e., 12 or 48 equivalents of hE3 or hE1, respectively (Fig. 3, C and D). We used the initial velocity of NADH formation, i.e., the last step in the PDHc reaction cycle, as a readout of overall hPDHc activity at saturating concentrations of the substrates pyruvate, CoA-SH, and NAD⁺. No indication for cooperative binding of hE1 or hE3 to hCore could be detected, and apparent dissociation constants (K_d)

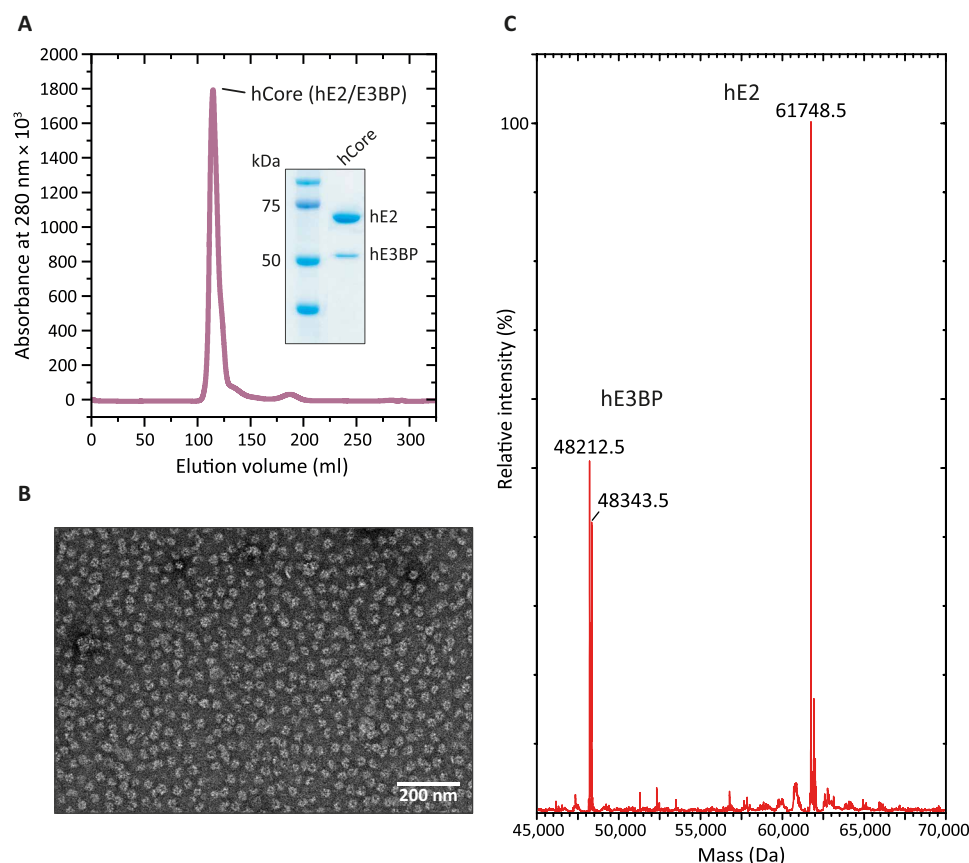


Fig. 1. Purification of the recombinantly produced hCore complex. (A) Size exclusion chromatogram of the final purification step and the corresponding SDS-polyacrylamide gel electrophoresis (PAGE) gel. Bands correspond to the first, void-volume peak. (B) Representative NS micrograph showing purified hCore particles. (C) Mass spectrum (relative intensity versus mass) of the purified hCore sample, confirming the identity of hE2 and hE3BP and the completeness of their lipoylation. Fully lipoylated hE2 (one lipoamide in each of the two lipoyl domains) uniformly contained an N-terminal methionine, while fully lipoylated hE3BP (one lipoamide in its single lipoyl domain) proved to be a mixture of chains with (48,343.5 Da) and without (48,212.5 Da) the N-terminal methionine.

were determined to be 1.5 ± 1.3 nM and 0.45 ± 0.18 nM for association of hCore with hE1 and hE3, respectively (Fig. 3C). Fitting the data with binding stoichiometry as an open parameter yielded 53.2 binding sites for hE1 and 9.1 binding sites for hE3, in good agreement with the hE2:hE3BP ratio of 48:12 within experimental error (Fig. 3D).

Next, we performed activity assays under conditions where hCore was not saturated with either peripheral subunit and the concentrations of hE1 and hE3 were varied simultaneously (fig. S5). We observed that the maximum turnover number of ~ 700 s⁻¹ was only obtained at excess concentrations of hE1, while hCore mixed with 48 hE1s and 12 hE3s reached $\sim 70\%$ of this value. On the one hand, the results indicated that free hE1 proteins further increased the measured PDHc activity when hCore was fully saturated with hE1 and pyruvate likely because hE1 catalyzes the rate-limiting reaction in the hPDHc cycle. On the other hand, about 12 equivalents of hE3 dimers were sufficient for achieving maximum activity, and excess, free hE3 dimers did not further increase activity. Together, these results are fully consistent with the 48:12 model of the hPDHc core complex and indicate that the observed maximum turnover number of 700 s⁻¹ at excess of hE1 may be a slight overestimation and that the true k_{cat} value of hPDHc, fully saturated with peripheral subunits and substrates at pH 7.4 and 25°C, is in the range of ~ 600 s⁻¹. This

value translates into a k_{cat} of ~ 12.5 s⁻¹ per hE1 heterotetramer, which catalyzes the rate-limiting step of the PDHc reaction cycle.

Next, we recorded the kinetics of hE1 and hE3 association with hCore via the increase in light scattering signal (fig. S6A). The kinetics, recorded under varying initial concentrations, could be well-fitted globally to apparent second-order reaction kinetics (fig. S6, C and D). The binding of both hE1 and hE3 to hCore proved to be notably fast, with on-rates (k_{on}) of 1.1×10^7 and 1.6×10^7 M⁻¹ s⁻¹, respectively. With the measured apparent K_d values, off-rates (k_{off}) of 0.04 s⁻¹ for hE1 and 0.007 s⁻¹ for hE3 were calculated. Furthermore, using the decrease in FAD fluorescence in hE3 upon binding to hE3BP (fig. S6B), we compared hE3 binding to free hE3BP monomers with binding to hE3BP in the context of hCore (fig. S6, E and F). The rate constant of hE3 association with hCore was similar to that determined with light scattering (2.4×10^7 and 1.6×10^7 M⁻¹ s⁻¹, respectively) but clearly lower than that of hE3 binding to free E3BP monomers (1.0×10^8 M⁻¹ s⁻¹). Still, hE3 binding to E3BP within hCore is a very rapid reaction, only about two orders of magnitude below the diffusion limit.

NS-EM analysis of hCore complexed with hE3

To analyze the spatial distribution of hE3BP subunits within the core, we performed an NS-EM single-particle analysis of hCore saturated

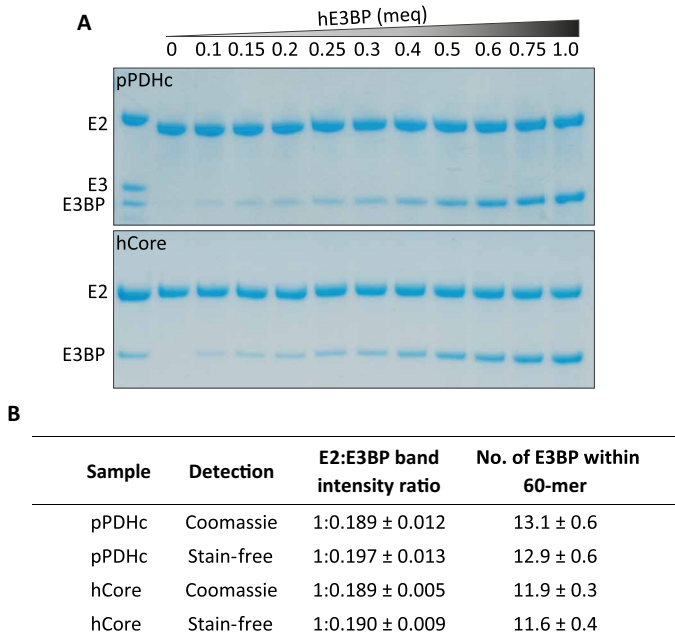


Fig. 2. Gel densitometry analysis for the porcine and human PDHc cores. (A) SDS-PAGE gels for pPDHc and hCore, and defined mixtures of purified hE2 and hE3BP (gels stained with Coomassie). Molar equivalents (meq) of hE3BP relative to the hE2 monomer are indicated above each lane. (B) Summary of the E2:E3BP band intensity ratios and the corresponding number of E3BP chains per 60-mer. The given standard errors were determined from three independent gels.

with hE3. As hE3 readily dissociated from hCore under the conditions of NS, we used GraFix fixation (36, 37), where core complexes incubated with excess hE3 were ultracentrifuged through a sucrose gradient in the presence of glutaraldehyde as cross-linker. Approximately 140,000 particles extracted from ~2000 micrographs were subjected to 2D classification (Fig. 4, A and B). As a negative control, we used glutaraldehyde-cross-linked hE2 homo-60-mer, which is unable to bind hE3. The obtained 2D class averages revealed conformational heterogeneity in the distribution of hE3 around hCore, most likely due to incomplete cross-linking and partial dissociation, but clearly without clustering of bound hE3 on one side of hCore as proposed earlier (Fig. 4C, left) (31). As a consequence of the unstructured 51-residue linker between the C-terminal, core-forming domain of hE3BP and its E3-binding domain (allowing for multiple orientations of bound hE3), the additional density of bound hE3 remained faint. Nevertheless, the NS-EM data were consistent with a uniform distribution of hE3BP chains within hCore and did not provide evidence for the clustering of hE3BPs within hCore.

To address potential cross-linking artifacts, we performed a control experiment in which hE2 homo-60-mer (lacking hE3BP) was subjected to GraFix in the presence of hE3. No additional density around the hE2 complex was observed in the control sample, excluding unspecific cross-linking between hE2 subunits and hE3 (Fig. 4C, right).

Cryo-EM structural analysis of the hE2 homo- and hCore hetero-oligomers

With the goal of obtaining high-resolution structural information on hCore, we first solved a single-particle cryo-EM structure of the

full-length hE2 homo-60-mer to a resolution of 3.3 Å (Fig. 5A). This established an efficient workflow for subsequent structural analysis of hCore and served as a direct reference for evaluating structural data on the hCore particle. Both complexes, hE2 60-mer and hCore, were prepared in the same manner and subjected to GraFix, as an initial screening of non-cross-linked samples resulted in a large number of broken particles where pentagonal faces (more than five trimers) had separated from the 60-mer, leaving the complexes incomplete or fully dissociated. The samples were subsequently frozen on holey grids via the “floating carbon” approach, wherein a 1.0- to 1.5-nm-thick carbon film was floated on the samples (38, 39). This freezing technique proved highly advantageous as it further improved stability against dissociation and remedied the clustering of particles.

As our biochemical data strongly supported the 48:12 hE2:hE3BP stoichiometry of hCore and the NS-EM analysis indicated a uniform distribution of hE3BP chains in the 60-mer (Fig. 4), we chose a tetrahedral arrangement of the 12 hE3BP copies in hCore, which is the only symmetry model fulfilling these criteria. Consequently, we analyzed 3D reconstructions of the hE2 homo-60-mer and the hCore particles by imposing tetrahedral symmetry.

The hE2 60-mer adopts icosahedral symmetry, with 20 identical hE2 trimers associating to a uniform dodecahedron. Although we solved the cryo-EM structure of the core-forming catalytic domains (hE2 residues 417 to 647) in the context of the full-length hE2 60-mer to 2.9-Å resolution by imposing icosahedral symmetry, we also obtained a 3D reconstruction at 3.3-Å resolution from the same dataset when we applied tetrahedral symmetry (Fig. 5A and fig. S7B). The latter map provided a good reference for the hCore map (obtained with imposed tetrahedral symmetry) and was used to minimize potential artifacts when interpreting the data on hCore (Fig. 5B and fig. S7B). In our analyses, we used full-length complexes; however, the N-terminal segments of hE2 and hE3BP containing lipoyl domains and peripheral subunit binding domains could not be resolved, as expected from the highly flexible linkers connecting these domains to each other and to the core-forming, C-terminal domains of hE2 and hE3BP (fig. S1A). Following the same refinement procedures (see Methods), the hE2 dataset yielded high-quality tetrahedral and icosahedral 3D reconstructions that were practically indistinguishable, confirming a uniform structure of the hE2 60-mer. In contrast, the hCore map, solved to an estimated global resolution of 3.4 Å when imposing tetrahedral symmetry (Fig. 5B), appeared substantially different from its icosahedral 3D reconstruction (which resembled that of the hE2 60-mer), agreeing with the hetero-oligomeric nature of hCore. The local resolution varied along the map with certain trimers better resolved than others. The overall hCore map exhibited lower quality than the hE2 3D reconstruction, as the cryo-EM densities appeared radially elongated (Fig. 5B). We attributed this behavior to multiple conformational states, which had been identified during data processing where reconstructed hCore particles differed in their dimensions (movie S1). We refer to this range of states as the “breathing” of the complex (40–42). Notably, the mode of breathing observed for the 3D reconstructions of hE2 and hCore differed. While the hE2 60-mer appeared to expand and shrink isotropically in tetrahedral symmetry, the hCore complexes changed their dimensions nonuniformly. In the symmetric, tetrahedral 3D reconstruction of hCore, certain parts of the map moved closer to the core center, while others simultaneously moved away. This difference between the two 3D

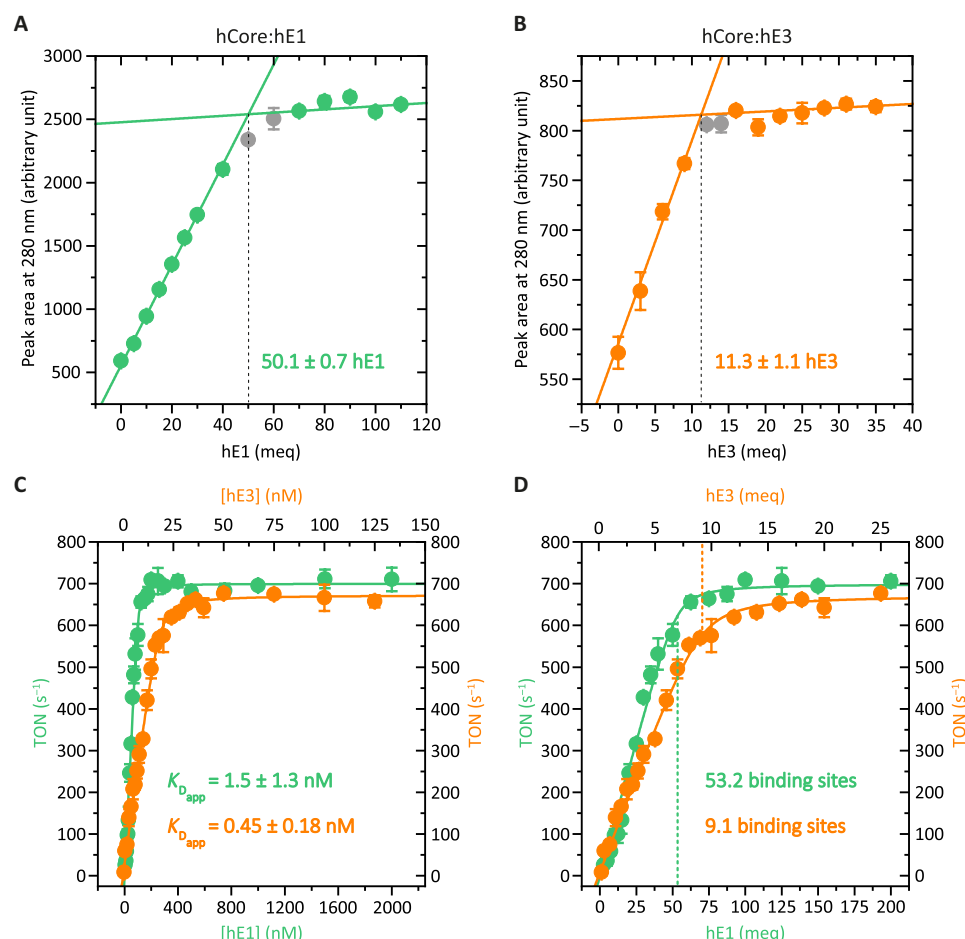


Fig. 3. Stoichiometry and affinity of binding of the peripheral subunits hE1 and hE3 to hCore, analyzed by analytical SEC and enzymatic activity. (A) Titration of hCore (80 nM) with hE1 (heterotetramers). Areas of the void volume peak, corresponding to the mixture of free and occupied hCore complexes, were plotted against the molar equivalents (meq) of hE1 added. Saturation is observed at 50.1 hE1 meq (hE1 heterotetramers to hCore, dashed line). (B) Titration of hCore with hE3 [same conditions as in (A)]. Saturation occurred at 11.3 meq of hE3 (hE3 dimers to hCore, dashed line). The equivalence points in (A) and (B) were obtained from the intersections of the linear fits (solid lines) of the data before and after the equivalence points. Data points near the equivalence points that were excluded from the linear fits are depicted in gray. (C) Dependence of the observed turnover numbers (TON) of hCore complexes (2 nM) saturated with 48 equivalents of hE1 on the concentration of hE3 added (orange data points) and of hCore complexes (2 nM) saturated with 12 equivalents of hE3 on the concentration of hE1 added (green data points). Data were fitted to a noncooperative binding equilibrium with the binding stoichiometry and dissociation constant as open parameters. (D) The same dataset as in (C), with meq instead of concentrations of hE1 and hE3. Dashed vertical lines indicate the number of binding sites fitted in (C).

reconstructions points to the incorporation of hE3BP as a key factor affecting complex dynamics.

To analyze the symmetrized hCore map, we searched for an asymmetric unit (AU) of hCore covering the density of five polypeptide chains with the hE2:hE3BP ratio of 4:1 (Fig. 5C). Substitution of a single hE2 chain in the AU with an AlphaFold model of hE3BP, subsequent model refinements against the map, and application of tetrahedral symmetry resulted in five potential structural models that fulfill the stoichiometry and symmetry requirements of hCore (Fig. 5D). Incorporation of hE3BP at positions 1 or 5 of the AU results in analogous hCore models, rotated by 120°. Models 1/5 correspond to an assembly of 4 hE3BP homotrimers and 16 hE2 homotrimers and exclude the occurrence of hE2-hE3BP heterotrimers. In contrast, a single hE3BP at positions 2, 3, and 4 of the heteropentameric AU results in models in which 12 heterotrimers consisting of two hE2s and one hE3BP assemble with 8 hE2

homotrimers. These models differ in that hE3BP at positions 2 and 4 of the AU allows for hE2-hE2 and hE2-hE3BP intertrimer interfaces, while the model with hE3BP at position 3 excludes hE2-hE3BP intertrimer interfaces and predicts that only hE3BP-hE3BP and hE2-hE2 intertrimer contacts exist.

We tested each of the five potential models for the best fit to the hCore cryo-EM map. Because of the considerable resemblance between hE2 and hE3BP (fig. S1), distinguishing between hE2 and hE3BP monomers in hCore proved challenging. The sequence identity and similarity of the hE2 and hE3BP C-terminal, core-forming domains resolved in our map are 48 and 69%, respectively, with the overall fold and secondary structures predicted to be highly similar for the two domains (fig. S1C). Consequently, we focused on identifying differences between the reconstructions of the hE2 60-mer and hCore. These ultimately pointed to model 3 as the most probable structure, in which the 12 hE2₂-hE3BP heterotrimers form six

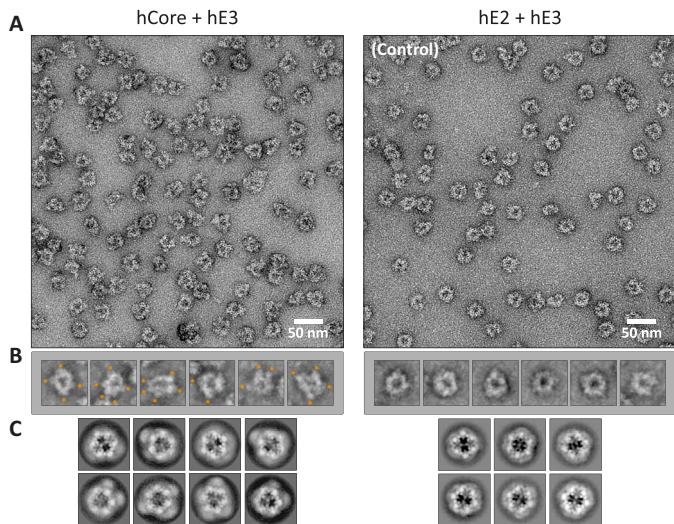


Fig. 4. NS-EM single-particle analysis of the hE2 homo- and hCore (hE2/E3BP) hetero-oligomers cross-linked with glutaraldehyde at excess hE3. (A) Representative NS micrographs recorded for both samples. In contrast to the hE2 60-mer, hCore complexes appear to be surrounded by extra density, attributed to hE3. (B) Examples of particles extracted during the data processing. Faint extra densities can be observed around the hCore complexes (indicated with orange asterisks). (C) 2D class averages generated from the two datasets (same scale). While a small halo around the hE2 60-mer likely corresponds to the N-terminal regions (lipoyl domains, E1-binding domains, and linkers) that became less disordered due to cross-linking, hCore complexes are clearly surrounded by extra densities, indicative of bound hE3.

hE3BP-hE3BP intertrimer contacts, while no intertrimer hE2-hE3BP contacts occur. Specifically, we noticed an unexpected density connecting two neighboring trimers for all such interfaces in the hE2 map, while the corresponding density was reduced or absent at some intertrimer interfaces in the hCore reconstruction (Fig. 5, E and F). On the basis of findings by Salem *et al.* (43) (crystallographic analysis of cross-linked barnase protein), we attributed this extra density to glutaraldehyde cross-linking of R471 located within the loop between helices H2 and H3 of one hE2 trimer with K490 from helix H3 in a neighboring hE2 trimer (Fig. 5F). Despite the low reactivity of the guanidinium group of Arg, glutaraldehyde cross-linking of Arg and Lys side chains has been reported previously and leads to a ring formation within the cross-link, for which we could detect a bulky density. This characteristic behavior is observed for all the twofold-related interfaces of the hE2 60-mer (see the top and bottom left panels in Fig. 5F for two examples). For the map of hCore, in contrast, the density connecting chains 1–2 and 4–5 in the AU is similar (Fig. 5F, bottom right), but a weaker signal is seen at the site of the predicted hE3BP-hE3BP pair visualized in model 3 (Fig. 5F, top right). Here, although the density that would correspond to R471 of hE2 is absent, indicating the presence of an hE3BP subunit, residue Q342 of the predicted hE3BP in the neighboring trimer occupies part of the density. It appears, however, that this density is not fully accounted for by Q342 of hE3BP and could also accommodate K490 of hE2. This observation, along with the density for helix H2 in the hCore map more closely resembling hE2 rather than hE3BP (enough signal to accommodate three-residue longer H2 helix of hE2), led us to conclude that misaligned or misclassified particles

during 3D classifications/3D refinements hampered unambiguous interpretation of the results, and the unusual glutaraldehyde cross-links between hE2s from different trimers did not provide enough differences to allow for the correct particle alignment and determination of Euler angles in the single-particle analysis of hCore.

To mitigate this problem, we revised our data processing strategy and used signal subtraction, followed by focused 3D classification (fig. S7A). The particles aligned in the initial 3D classification and subsequent local refinement were subjected to signal subtraction to preserve the predicted hE3BP-hE3BP pair while removing the rest of the particles. This pool was then used in the 3D classification, where classes showing features characteristic of hE3BP (no cross-linking between hE3BPs and reduced density at H2 helix) were selected, reverted to their original particles, and locally refined. This procedure was repeated and resulted in a final pool of about 144,000 particles. Removal of particles upon 3D classifications and selection of class averages caused a decline in the quality of the reconstruction, which generated a map at 3.7-Å global resolution in the last local refinement (Fig. 6A and fig. S7B). The local resolution of this 3D reconstruction varied across the complex: Interior parts were estimated at ~3.4-Å resolution, while the outer region was less resolved and reached ~5-Å resolution. Model 3 (Fig. 5D) was refined against the map and the structure is shown in Fig. 6B. According to this model, the hCore complex adopts tetrahedral symmetry, with 12 hE3BPs forming six hE3BP-hE3BP intertrimer (between two hE2-hE3BP heterotrimers) interfaces along the twofold rotational symmetry axes. The only other mode of intertrimer interactions involves two hE2s, as no hE2-hE3BP contacts between the neighboring trimers are formed in this model. Here, the density for glutaraldehyde cross-linking between R471 and K490' is present between the two chains predicted to be hE2s (Fig. 6C, top), while no such density appears between the hE3BPs (Fig. 6C, bottom). Furthermore, the H2 helices involved in intertrimer contact formation show clear differences between the sites predicted to be occupied by hE2 or hE3BP (Fig. 6C, right). More density is observed for hE2s as its H2 helix is three residues longer than the equivalent helix in hE3BP. The loop connecting helices H2 and H3 is also less constrained, as the three additional amino acids allow for a more relaxed transition between the two helices. In contrast, the density for the H2 helix resolved at the presumed hE3BP location is slightly shorter, making it unlikely to accommodate the extra residues of hE2.

In summary, our cryo-EM analysis supported model 3 for the general architecture of the hCore complex. In the structure illustrated in Fig. 6B, 12 hE3BP molecules (six hE3BP pairs) form six symmetric interfaces between neighboring trimers, which are held together by a network of contacts. These are primarily hydrophobic in nature and include hE3BP residues V315, L319, F331, A383, G385, I386, P497, I498, L500, and A501 (Fig. 6D).

Analysis of the hE2 60-mer and hCore with native and cross-linking mass spectrometry

We next used MS to obtain independent evidence for the correct hCore structure. First, the hE2 60-mer and hCore were subjected to native MS analysis to identify the oligomeric building blocks of the two assemblies. Both samples proved to be prone to spontaneous dissociation under native MS conditions, preventing the detection of complete complexes or larger oligomers. However, trimeric building blocks could be observed for both particles. Comparison of the hE2 homo- and hCore hetero-oligomers revealed clear differences

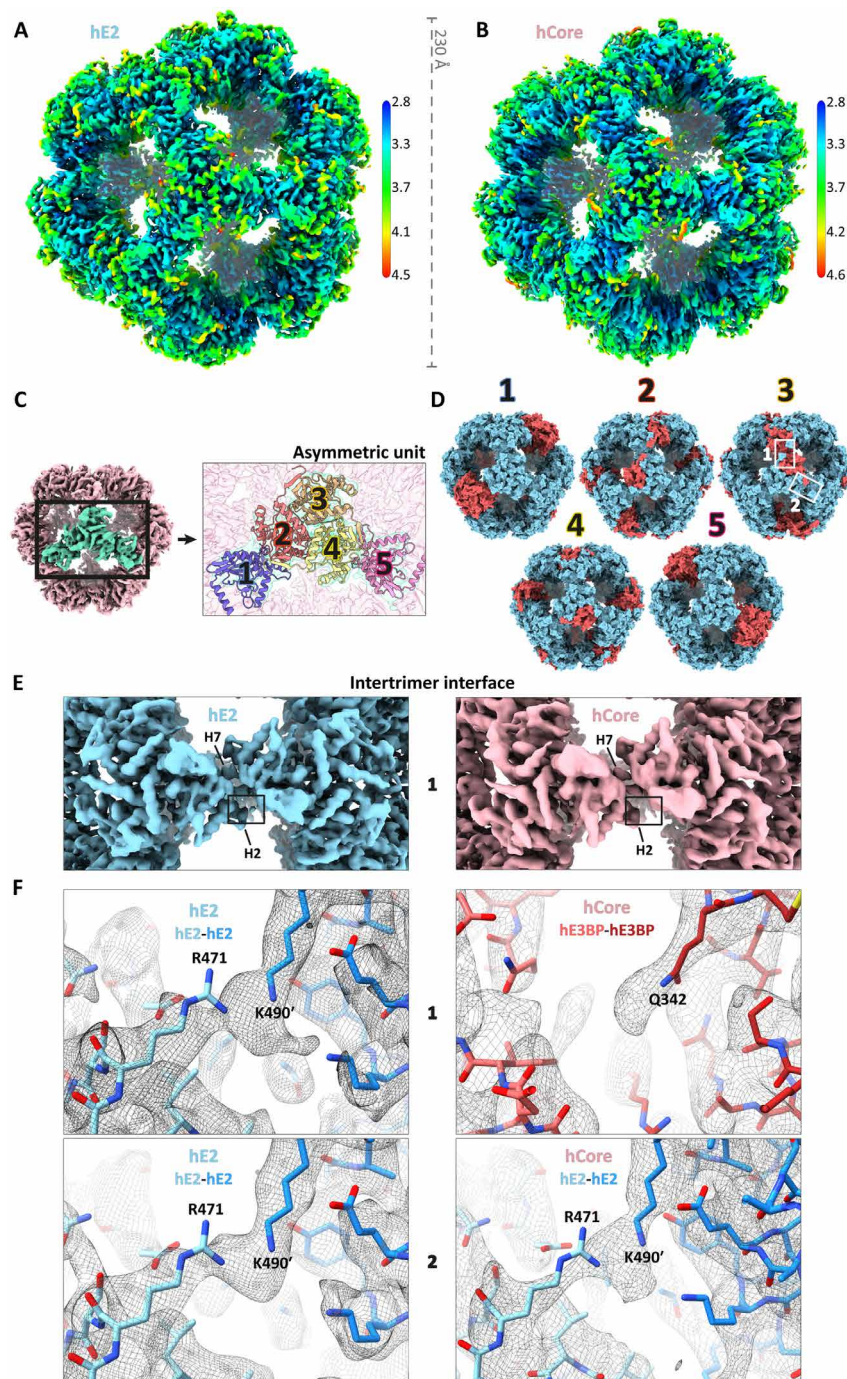


Fig. 5. Cryo-EM analysis of the hE2 homo- and hCore (hE2/E3BP) hetero-oligomers. (A) Cryo-EM map of the hE2 60-mer solved by imposing tetrahedral symmetry. The map is colored according to local resolution. (B) Cryo-EM map of hCore, refined with applied tetrahedral symmetry and colored according to local resolution. (C) (AU; tetrahedral symmetry) of the hCore map. Each chain of the demonstrative hE2 homopentamer fitted into the AU that could be potentially substituted with hE3BP is numbered. (D) Five potential hCore models generated by applying a tetrahedral symmetry to five different heteropentamers consisting of four hE2s (blue) and one hE3BP (red). The numbers above models indicate which hE2 chain of the AU was substituted with hE3BP. For model 3, the hE3BP-hE3BP and hE2-hE2 intertrimer interfaces are indicated with boxes 1 and 2, respectively. (E) Map of an hE2-hE2 intertrimer interface in the hE2 60-mer (left) and of the predicted (model 3) hE3BP-hE3BP intertrimer interface in hCore (right). The helices H2 and H7 responsible for intertrimer contacts are labeled. (F) Intertrimer cross-link formation in the hE2 homo- and hCore hetero-oligomers. Close-up view of hE2-hE2 intertrimer interfaces in the hE2 60-mer and the hE2-hE2 and hE3BP-hE3BP intertrimer interfaces in hCore [interfaces 1 and 2, as indicated in (D)]. A continuous density between neighboring trimers presumably corresponds to the glutaraldehyde molecule cross-linking R471 and K490' of two hE2s via ring formation. A similar pattern is seen for the hE2 60-mer (bottom left) and the assumed hE2-hE2 interface within hCore (bottom right), indicating cross-linking. In the case that two hE3BPs met at the twofold interface, however, cross-linking would not be expected. Here, the assumed hE3BP-hE3BP interface lacked continuous intertrimer density (top right), whereas the density for cross-link formation is seen for the equivalent hE2-hE2 interface within the hE2 60-mer (top left).

in the recorded mass/charge ratio (m/z) spectra. Whereas hE2 60-mer gave distinct peaks for monomers, homodimers, and homotrimers of hE2 (fig. S8A), additional signals were detected for hCore (fig. S8B). Although these additional peaks could be attributed to the hE2-hE3BP heterodimers and hE2₂-hE3BP heterotrimers, sample complexity (overlapping peaks) and high noise made the assignment complicated. To verify the presence of hE3BP in the identified trimers, ions observed in the range m/z 6000 to 6300 were isolated by MS/MS and subjected to gas-phase collision-induced dissociation (CID) (fig. S9). The CID experiments confirmed that both hE2 and hE3BP are present in the trimers, as additional, hE3BP-specific signals were only observed in trimers from hCore. Subsequently, the resolution of the recorded peaks was improved by using a cyclic ion mobility MS (cIM-MS) experiment (Fig. 7 and fig. S10) (44). The separation of ions based on both their mobility and m/z ratios allowed us to further increase the splitting of the double peaks and confirm the presence of hE2₂-hE3BP heterotrimers.

Together, the native MS analysis demonstrated that both hE2 homo- and hE2₂-hE3BP hetero-trimers are present within hCore. This observation, along with the absence of hE3BP homotrimers, excludes models 1/5 (Fig. 5D) for hCore. It does not, however, allow discriminating between models 2/4 and 3 because all these models represent assemblies of 12 hE2₂-hE3BP heterotrimers and 8 hE2₃ homotrimers.

To get direct information to distinguish between models 2/4 and 3, we applied XL-MS (45) on the hCore complexes treated with bis(sulfosuccinimidyl) suberate (BS³), a homobifunctional reagent cross-linking amino groups of lysine residues. Digestion with trypsin after cross-linking and MS analysis of the tryptic peptides allowed us to identify 77 hE2:hE3BP, 71 hE2:hE2 (inter- and intramolecular), and 74 hE3BP:hE3BP (inter- and intramolecular) cross-links. Although the majority of the cross-linked lysines were located within the flexibly tethered E1-binding domain of hE2, the E3-binding domain of hE3BP, the lipoyl domains of hE2 or hE3BP, or unstructured linkers (fig. S11) and could not be used to distinguish between models 2/4 and 3, we obtained clear evidence supporting hCore model 3 from cross-links between the C-terminal core-forming domains of hCore (catalytic domain of hE2 and the pseudocatalytic domain of hE3BP). Most notably, we detected multiple hE3BP-hE3BP intertrimer cross-links that are diagnostic for model 3: These include cross-links between the lysine pairs K321:K321', K321:K488', K321:K491', and K394:K491' at the hE3BP:hE3BP' intertrimer interfaces of model 3, with approximate alpha carbon distances of 26, 17, 16, and 28 Å, respectively (Fig. 8, bottom right). The other detected cross-links between lysine pairs of the pseudocatalytic domain of hE3BP (K314:K460, K321:K450, K321:K460, K334:K379, K334:K384, and K341:K384) were consistent with both intertrimer and intra-hE3BP monomer cross-links (all alpha carbon distances below 26 Å). Of these, the pairs K341:K384, K321:K450, and K321:K460

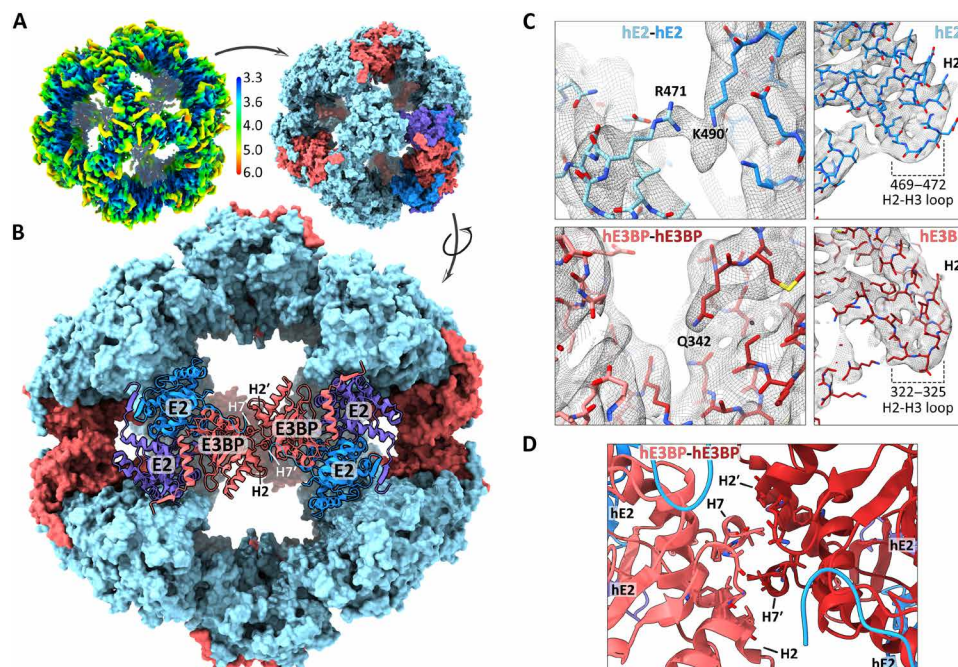


Fig. 6. The proposed model of the hCore complex. (A) Final map of the hCore complex colored according to local resolution estimates (left) and surface representation of the corresponding refined model (right). The proposed model shows the spatial distribution of hE2s [cyan, blue, or violet; see (B)] and hE3BPs (red) within a 60-subunit assembly. (B) Proposed model of hCore [same as (A)] viewed down the twofold rotational symmetry axis. The two neighboring hE2₂-hE3BP heterotrimers (cartoon representation) in the center of the image interact via contacts between helices H2 and H7 of adjacent hE3BPs (red). The hE2 subunits in these highlighted heterotrimers are colored blue and violet, while all other hE2 subunits are depicted in cyan. (C) Model fitting in the cryo-EM map shown in (A). The left panels show the putative hE2-hE2 and hE3BP-hE3BP intertrimer interfaces in hCore, equivalent to those shown in Fig. 5F. The density corresponding to cross-linked R471 and K490' is observed for the predicted hE2-hE2 interaction (top), whereas such density is absent from the putative hE3BP-hE3BP interface (bottom). The model fit is also demonstrated for the H2 helix where hE3BP is three residues shorter than hE2 (right). (D) The proposed hE3BP-hE3BP interface. Side chains of residues suggested to be involved in intertrimer contacts are shown in stick representation. The interaction is hydrophobic in nature and includes primarily residues of helices H2 and H7.

are also likely intertrimer cross-links. The alpha carbon distance of the K341:K384 pair is only ~ 10 Å in the refined model 3, and the latter two cross-links are particularly likely to form between hE3BPs of two heterotrimers considering their close proximity to K488' and K491' which are involved in intermolecular cross-links (Fig. 8, bottom right).

Models 2/4 and 3 also predict hE2-hE3BP cross-links between the core-forming domains of hCore. While only intratrimer hE2-hE3BP cross-links can occur in model 3, models 2/4 predict both intra- and intertrimer hE2-hE3BP cross-links. The top panels of Fig. 8 show the identified hE2-hE3BP cross-links that are consistent with both models 2/4 and model 3 (intertrimer versus intratrimer, respectively). Together, the XL-MS data and identification of multiple hE3BP-hE3BP intertrimer cross-links clearly support model 3 of hCore.

DISCUSSION

Despite their central role in energy metabolism and decades of research, the structural principles underlying the architecture and

exact subunit composition of mammalian PDHcs remained elusive. On the one hand, the flexible linkers between all domains of the mammalian PDHc core subunits E2 and E3BP, which are crucial for PDHc activity and allow transfer of catalytic intermediates between the active sites of E1, E2, and E3, prevent the structure determination of the completely assembled PDHc complex at atomic resolution. On the other hand, there had also been uncertainty about the stoichiometry of the fully assembled PDHc megaenzyme, and two different models for the pseudoicosahedral, hetero-60-meric PDHc core were proposed, with E2:E3BP ratios of 40:20 and 48:12. As E2 monomers exclusively interact with heterotetramers of the peripheral subunit E1, and E3BP monomers only interact with homodimers of the peripheral subunit E3, the E2:E3BP ratio dictates the stoichiometry of the fully assembled PDHc complex. Using recombinant human PDHc as a model, we could provide convincing evidence for the 48:12 model of the mammalian PDHc core, as calibrated gel densitometry experiments on human and porcine PDHc and titration of hCore with the recombinant subunits hE1 and hE3 consistently confirmed the hE2:hE3BP ratio of 48:12. An explanation for the previously reported hE2:hE3BP ratio of 40:20 could be the tendency of monomeric hE3BP to form large aggregates in the presence of substoichiometric amounts of hE2, and the potential copurification of hCore with aggregated hE3BP that could have led to an overestimation of the hE3BP content in hCore (fig. S12). Specifically, the coexpression of hE2 and hE3BP in *E. coli* with tunable levels of hE2 always yielded hE3BP aggregates when the hE3BP levels exceeded those of hE2. When the concentrations of hE2 and hE3BP were approximately even, only hetero-oligomeric hCore 60-mer formed. The 48:12 stoichiometry thus appears to be the preferred assembly state of hCore, despite the fact that the hE2 homo-60-mer is more stable against dissociation and unfolding than hCore (20). Consequently, the formation of hCore must be under kinetic control in vivo. The assembly of PDHc in vivo had been shown to be dependent on molecular chaperones (46), which might favor the formation of the thermodynamically less favored hCore particle. Notably, we could not reconstitute the hCore particle by mixing hE2 60-mer with hE3BP monomers in vitro (fig. S4B). The functionally equivalent Hsp60/Hsp10 and Hsp70 chaperones in the *E. coli* cytoplasm and the mitochondrial matrix (47) would thus be a plausible explanation of the efficient in vivo assembly of hCore in *E. coli* from coexpressed hE2 and hE3BP with reproducible 48:12 stoichiometry. Further support for the hypothesis that *E. coli* chaperones assisted the in vivo assembly of recombinant hCore comes from the observation that small amounts of *E. coli* Hsp70 (DnaK) were copurified with hCore (Fig. 7).

Because of the high pseudosymmetry of hCore and considerable similarity between the core-forming domains of hE2 and hE3BP, incorrect Euler angle determination hampered precise 3D classification and resulted in a 3D reconstruction where the more abundant hE2 (fourfold excess over hE3BP) might persist within the assumed hE3BP-hE3BP interface rendering the final cryo-EM map ambiguous. As we could not unambiguously visualize the exact location of the hE3BP subunits in hCore using cryo-EM, we cannot completely exclude that alternative, biologically relevant conformational states of hCore with 48:12 stoichiometry exist in addition to our proposed structural model 3 (see Figs. 5D and 6B). Nevertheless, we obtained multiple independent evidence showing that model 3 is the most plausible hCore structure and, if alternative architectures with 48:12 stoichiometry existed, the thermodynamically most stable (i.e., most

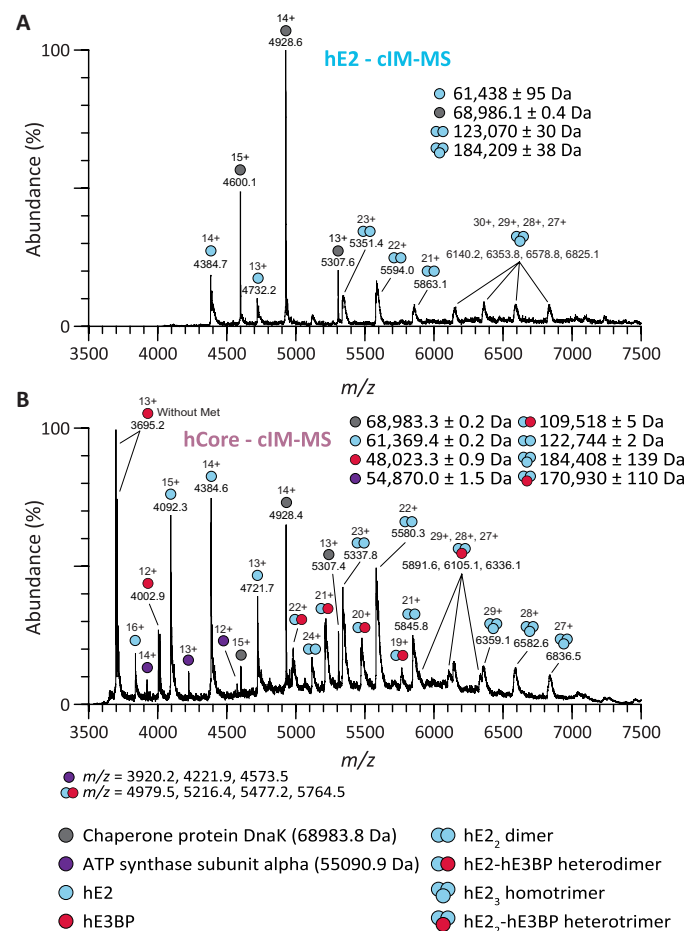


Fig. 7. Native mass spectrometry on the hE2 homo- and hCore (hE2/hE3BP) hetero-oligomers. Cyclic ion mobility mass spectrometry analysis of the hE2 (A) and hCore (B) complexes prepared in 200 mM ammonium acetate. The identified peaks are labeled with circles representing the oligomeric state of the species, as well as their respective m/z values and charge states.

populated) state of hCore. NS-EM data on hE3-saturated hCore were only consistent with an even (symmetrical) distribution of hE3BP within hCore. The measured 48:12 stoichiometry of pseudicosahedral hCore leaves a tetrahedral arrangement of the 12 hE3BP copies in the complex as the only possible symmetry. The even hE3BP distribution also provides a strong functional advantage over the previously proposed hCore model with hE3BP clustering (31), as the latter model would hinder shuttling of catalytic intermediates from active sites opposite to the hE3BP cluster in the 60-subunit core to the hE3BP-bound hE3 subunits. Native MS and XL-MS data excluded four out of five possible tetrahedral arrangements of hE3BP. Specifically, native MS excluded the existence of the

hE3BP homotrimers and hE2-hE3BP₂ heterotrimers and confirmed that hCore only consists of homotrimeric hE2₃ and heterotrimeric hE2₂-hE3BP building blocks. Regarding the precise *in vivo* assembly mechanism of hCore, this likely excludes an assembly model in which, e.g., an hE2 48-mer is first formed as a scaffold, followed by binding of 12 hE3BP monomers to this scaffold. Instead, the assembly of hCore likely proceeds via hE2₃ homotrimers and hE2₂-hE3BP heterotrimers as obligatory assembly intermediates.

In an attempt to determine the hCore architecture by single-particle cryo-EM, simultaneous processing of the hE2 homo- and hCore hetero-oligomers allowed us to observe clear differences between the two maps: Incorporation of hE3BPs altered core dynamics and affected its breathing motion and the 3D reconstruction (Fig. 5 and move S1). The potentially weaker hE3BP-hE3BP intertrimer interactions compared to hE2-hE2 intertrimer contacts, possibly caused by the shorter helices H2 and H7 in hE3BP, might allow the core to adopt multiple conformations that could favor substrate channeling between the active sites of hPDHc. The different dynamics of oligomeric hE2 and hCore could explain why Hiromasa *et al.* (32) observed a larger excluded volume for hCore complexes compared to hE2 60-mer measured by small-angle x-ray scattering. Furthermore, following an analogous analysis as that performed by Jiang *et al.* (23), we used PDBEPIA (48, 49) to estimate the solvation free energy gain (Δ^1G) upon formation of the two types of intertrimer interfaces in our refined model. This predicted Δ^1G values -17.2 and -8.4 kcal/mol for the hE2-hE2 and hE3BP-hE3BP interfaces, respectively. The more stable hE2-hE2 interfaces should also favor the hE2:hE3BP stoichiometry of 48:12 relative to the previously suggested 40:20 ratio.

The differences between the cryo-EM maps of the hE2 homo- and hCore hetero-oligomers are also fully consistent with model 3 of hCore, composed of 12 hE2₂-hE3BP heterotrimers and 8 hE2₃ homotrimers, in which 12 hE3BPs form six intertrimer hE3BP-hE3BP interfaces along the twofold rotational symmetry axes (Figs. 7 and 8) and are arranged according to tetrahedral symmetry (Fig. 6). The other type of intertrimer contacts in model 3 includes 24 hE2-hE2 pairs (24 intertrimer interfaces left), thereby excluding potential twofold-related hE2-hE3BP interactions.

In conclusion, the presented biochemical data acquired from gel densitometry, SEC, and enzymatic activity assays provide evidence for a 48:12 stoichiometry of the human PDHc core, in full accordance with the core of native porcine PDHc. The stoichiometry of hCore predicts that 48 hE1 heterotetramers and 12 hE3 homodimers are bound to hCore when hPDHc is fully saturated with the peripheral subunits, corresponding to a total mass of 12.0 MDa with bound cofactors. Moreover, our single-particle EM and MS analyses allowed us to propose a tetrahedral model for the architecture of the core, with an even distribution of hE3BP. Solutions toward the goal of determining a high-resolution structure of hCore in which hE2 and hE3BP can be clearly distinguished might require the modification of the hE3BP subunits with a label of sufficient mass to ensure the correct particle alignment and determination of Euler angles. In addition, a detailed cross-linking-MS analysis of the fully assembled hPDHc could potentially provide additional information on the hCore architecture and confirm the positions of hE3BP subunits in our proposed model.

During the revision of this manuscript, a preprint article by Wang *et al.* (50) was published, describing the structural characterization of PDHc complexes from porcine heart mitochondria *in situ* using

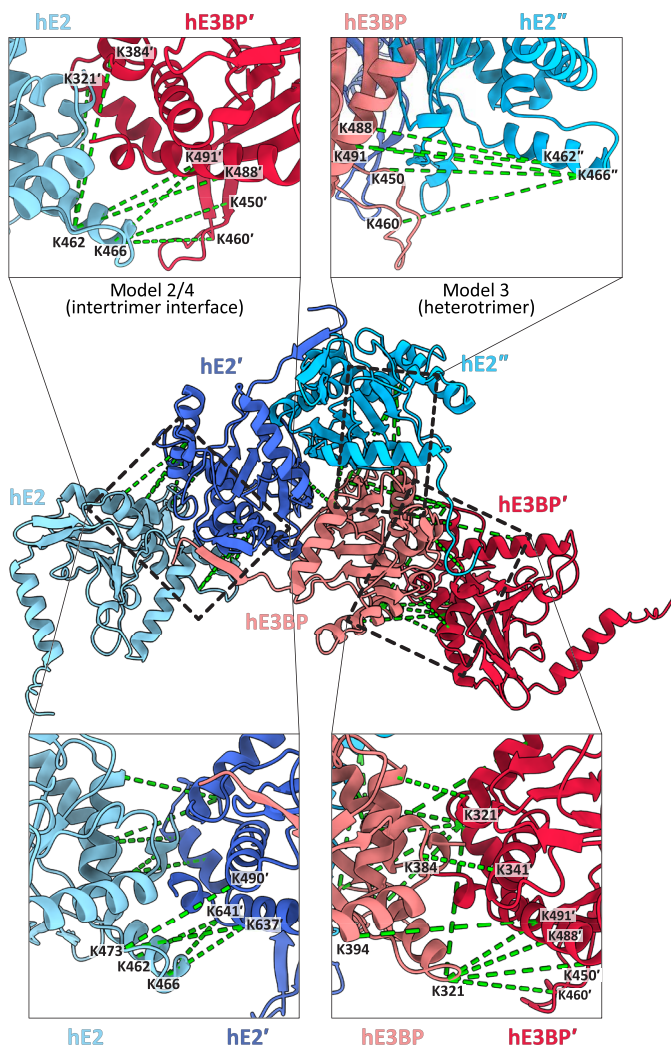


Fig. 8. Cross-linking mass spectrometry analysis of hCore. Identified BS³ cross-links (green dashed lines) were mapped onto a heteropentamer consisting of three hE2s and two hE3BPs that had been extracted from the hCore model 3 (Fig. 6B). The relevant cross-links are highlighted in the bottom panels, showing a magnified view of the intertrimer hE2-hE2 and hE3BP-hE3BP interfaces. Lysine residues forming cross-links at one side of the symmetric interface are labeled. The top panels show identified cross-links that are consistent with both model 3 (right) and models 2/4 (left). The hE2-hE3BP cross-links 462:321 and 462:384 (top left) are likely artifacts of XL-MS, as both cross-links would disrupt the intertrimer interface.

cryo-electron tomography. The authors confirmed the 48:12 stoichiometry of the porcine PDHc core complex by showing that the maximum observed occupancy with peripheral subunits in vivo was 46 E1 heterotetramers and never exceeded 12 E3 dimers. The results were also consistent with an even distribution of E3BP within the porcine PDHc core particle. Notably, the occupancy of the core particles proved to be highly variable, with an average occupancy of 21 E1 heterotetramers and only 4 E3 dimers. The results demonstrate that variable occupancy of the PDHc core with peripheral subunits adds a level of complexity to the PDHc structure and function. Extrapolation of our recorded activity profiles of human PDHc at variable hE1 and hE3 occupancy in vitro (fig. S5) to porcine PDHc predicts that the enzyme only reaches about 20% efficiency in porcine heart mitochondria compared to its activity at full occupancy with peripheral subunits. PDHc activity in different tissues might thus be regulated, in addition to phosphorylation/dephosphorylation of E1 (24–27), via the occupancy of the core with E1 and E3.

MATERIALS AND METHODS

Protein production and purification

Codon-optimized genes (without mitochondrial transit peptide) encoding human E1 α (PDHA1), E1 β (PDHB), E2 (DLAT), E3BP (PDHX), and E3 (DLD) for expression in *E. coli* were synthesized (GeneArt, Thermo Fisher Scientific) and cloned into pET-21a vectors (T7 promoter, lac operator) either via ligation of polymerase chain reaction products or Gibson assembly (primers synthesized by Microsynth). The genes encoding hE1 chains α and β , separated by a ribosome-binding site, were coexpressed as dicistronic operon. An analogous coexpression construct was generated for hE2 and hE3BP in addition to plasmids for the production of the individual subunits. In the case of hE1 α , an amino-terminal Strep-tag II followed by a tobacco etch virus (TEV) protease cleavage site was introduced. The hE2 gene encoded an N-terminal His₆ tag and a subsequent TEV cleavage site, primarily for obtaining higher expression yields. To allow tunable expression of hE2, its DNA sequence was inserted into pAC-P_{tetT7}-HIV (51) by Gibson assembly (in between NdeI and SpeI restriction sites) to give pAC-P_{tetT7}-hE2. Plasmids from transformed NEB 5-alpha cells (New England Biolabs) were purified with the NucleoSpin Plasmid kit (Macherey-Nagel). All plasmids were verified by DNA sequencing (Microsynth).

All proteins were recombinantly produced in LB(Miller)-grown *E. coli* BL21 (DE3) cells, except for hE2/E3BP coexpression, where BL21 Star (DE3) (Invitrogen) cells were used. Gene expression under the control of the *lac* repressor was induced at OD₆₀₀ ~ 0.8 with 1 mM isopropyl- β -D-thiogalactopyranoside. Bacteria were then further grown for 16 (hE1 and hE3) or 20 hours (hE2, hE3BP, and hE2/E3BP), at 20° or 25°C (hE1). Cultures producing hE3 were supplied with 0.25 mM FAD, while lipoic acid (3 mM) was added to the medium of hE2-, hE3BP- and hE2/E3BP-producing cells (lower lipoic acid concentrations in the medium only yielded mixtures of partially lipoylated proteins).

The initial steps of purification were similar for all proteins used in this study. Harvested cells were resuspended in lysis buffer [20 mM Hepes-NaOH (pH 7.4), 150 mM NaCl, 5 mM CaCl₂, and 5 mM MgCl₂] supplied with 1 mM phenylmethylsulfonyl fluoride (PMSF), deoxyribonuclease (DNase I; 50 U/ml), and 1 $\times$ protease inhibitor cocktail (cOMplete, Roche). Bacteria coexpressing hE1 α and β chains were resuspended in hE1 lysis buffer [100 mM tris-HCl

(pH 8.0), 150 mM NaCl, 1 mM EDTA, 1 mM dithiothreitol (DTT), and 2 mM ThDP]. Bacteria were lysed with an M110-L microfluidizer (Microfluidics). Lysates were centrifuged in an SS-34 rotor (Thermo Fisher Scientific) for 45 min at 48,000g. Proteins were then purified from the soluble fraction of the cell extract according to the following protocols.

Lysate containing Strep-tagged hE1 was loaded onto a Strep-Tactin XT Superflow high-capacity affinity resin (IBA Lifesciences), and hE1 was eluted with biotin according to the manufacturer's instructions. Fractions containing hE1 were confirmed via SDS-PAGE gel, pooled, and loaded onto a HiLoad 26/600 Superdex 200 column (Cytiva) equilibrated with PB buffer [20 mM Hepes-NaOH (pH 7.4), 140 mM KCl, 10 mM NaCl, and 1 mM MgCl₂]. Fractions containing pure hE1 were verified with SDS-PAGE, pooled, concentrated, flash-frozen in liquid nitrogen, and stored at –80°C.

Lysate containing hE3 was dialyzed against buffer A [20 mM Hepes-NaOH (pH 8.0), 4°C] and subsequently loaded onto a 20-ml Q Sepharose Fast Flow column (Cytiva). hE3 was eluted with a linear gradient from zero to 1 M NaCl in buffer B [20 mM Hepes-NaOH (pH 8.0) and 1 M NaCl, 4°C]. Fractions with hE3 were pooled and a saturated ammonium sulfate solution was added dropwise to 25% saturation (52). This sample was loaded onto a Phenyl Sepharose column (Cytiva) equilibrated with HIC buffer B [20 mM Hepes-NaOH (pH 8.0), 4°C, and 25% (NH₄)₂SO₄]. hE3 was eluted with a linear gradient of buffer A. Fractions with hE3 were pooled, brought to 150 mM NaCl with buffer B, concentrated, and incubated overnight with a 10-fold molar excess of FAD to ensure complete saturation with the cofactor. The hE3 sample was then loaded onto a HiLoad 26/600 Superdex 200 column equilibrated with PB buffer. Fractions containing pure hE3 were combined, concentrated, flash-frozen in liquid nitrogen, and stored at –80°C.

For the purification of hE3BP, the protein was precipitated from the soluble fraction of the cell extract with ammonium sulfate (35% saturation). After centrifugation (SS-34, 30 min, 48,000g, 4°C), the pellet was resuspended in buffer A and loaded onto a Q Sepharose Fast Flow column. hE3BP was eluted with a linear NaCl gradient (0 to 1 M). Fractions containing hE3BP were pooled, concentrated, and loaded onto a HiLoad 26/600 Superdex 200 column equilibrated with PB buffer. Fractions from the main peak with hE3BP monomers were combined, flash-frozen in liquid nitrogen, and stored at –80°C.

Lysates containing hE2 homo- or hCore hetero-oligomers were processed according to similar protocols that only differed in the ammonium sulfate concentrations for precipitation of hE2 and hCore from the lysates (30 and 25% saturation, respectively). Following centrifugation (SS-34, 30 min, 48,000g, 4°C), the pellets were resuspended in PB buffer supplemented with 1 mM DTT. After ultracentrifugation [MLA-80 (Beckman Coulter), 35 min, 60,000 rpm, 4°C], the supernatants were subjected to density gradient centrifugation [10 to 30% (w/v) sucrose, SW 32 rotor (Beckman Coulter), 24,000 rpm, 16 hours, 4°C]. Fractions from the gradient were analyzed by SDS-PAGE. Fractions containing hE2 60-mer or hCore were combined, concentrated, and loaded onto a HiLoad 26/600 Superdex 200 column equilibrated with PB buffer. Fractions with pure protein were combined, flash-frozen, and stored at –80°C.

Determination of protein concentrations

The concentrations of the purified proteins were determined via their specific absorbance at 280 nm. The following molar extinction

coefficients (ϵ_{280}) were used: $\epsilon_{280}(\text{hE1 heterotetramer}) = 157,921 \text{ M}^{-1} \text{ cm}^{-1}$; $\epsilon_{280}(\text{hE3 homodimer}) = 61,530 \text{ M}^{-1} \text{ cm}^{-1}$; $\epsilon_{280}(\text{hE2 60-mer}) = 2,312,793 \text{ M}^{-1} \text{ cm}^{-1}$; $\epsilon_{280}(\text{hE3BP}) = 20,970 \text{ M}^{-1} \text{ cm}^{-1}$; and $\epsilon_{280}(\text{hCore}) = 2,250,911 \text{ M}^{-1} \text{ cm}^{-1}$.

For hE3, the impact of FAD absorbance at 280 nm was considered. The concentrations of hE2 60-mer and hCore were corrected for the contribution of light scattering as described (53) by plotting $\log(\text{absorbance})$ versus $\log(\text{wavelength})$, extrapolating the linear region of this plot (310 to 340 nm) back to 280 nm and subtracting this extrapolated value from the measured absorbance.

Purification of pPDHc

Porcine mitochondria were purified as previously described (54). Frozen mitoplasts (30 g) were thawed in PB lysis buffer [20 mM Hepes-NaOH (pH 7.4), 140 mM KCl, 10 mM NaCl, 1 mM MgCl_2 , 2 mM DTT, DNase I (100 U/ml), 2 mM PMSE, and 2 $\times$ protease inhibitor cocktail] and brought to a volume of 100 ml before homogenization (Homogenizer HG-15D, Witeg). Then, 20 ml of SB buffer [20 mM Hepes-NaOH (pH 7.4), 140 mM KCl, 10 mM NaCl, 1 mM MgCl_2 , and 9.6% (v/v) Triton X-100] were added dropwise along with 6 mg of ribonuclease A for membrane solubilization and degradation of ribosomes. The solution was then stirred for 30 min at 4°C and centrifuged (SS-34, 20 min, 48,000g, 4°C). The supernatant was subjected to ultracentrifugation in a Type 45 Ti rotor (Beckman Coulter, 40,000 rpm, 3 hours, 4°C) and the pellet was resuspended in 5 ml of PB buffer supplemented with 1 mM DTT. The cloudy suspension was cleared by centrifugation (Eppendorf 5425R, 30 min, 20,000g, 4°C) and the particles were once again sedimented by ultracentrifugation (MLA-80, 1 hour, 60,000 rpm, 4°C). The pellet was dissolved in 800 μl of PB buffer (+1 mM DTT), cleared (Eppendorf 5425R, 15 min, 16,000g, 4°C), and loaded onto a sucrose density gradient [20 to 40% (w/v) sucrose, SW 32.1 rotor (Beckman Coulter), 14 hours, 28,000 rpm, 4°C]. After gradient fractionation, every second fraction was tested for PDHc, OGDHc (α -ketoglutarate dehydrogenase complex), and BCKDHc (branched-chain α -keto acid dehydrogenase complex) activity by mixing 20- μl aliquots with a standard reaction mixture (see below) containing either pyruvate, α -ketoglutarate, or α -ketoisocaproate. Reactions were followed via the increase in NADH absorption at 340 nm in a plate reader (Synergy-2, Agilent Technologies). On the basis of enzymatic activity and the SDS-PAGE analysis, fractions containing maximum PDHc and minimum OGDHc activity were combined and subjected to a two-step PEG precipitation. A 12% PEG-6000 solution (v/v, 50% w/v stock) was added and the mixture was incubated on ice for 5 min. After centrifugation (Eppendorf 5425R, 15 min, 16,000g, 4°C), the PEG-6000 concentration in the supernatant was increased to 15% (v/v), and the solution was centrifuged again following a 5-min incubation. The pellet fraction was resuspended in PB buffer. SDS-PAGE and PDHc and OGDHc activity in the supernatant and resuspended pellet fractions showed that the dissolved pellet from the 15% (v/v) PEG step only contained pPDHc, which was flash-frozen and stored at -80°C .

Gel densitometry analysis

Gel densitometry analyses were performed for purified pPDHc and hCore. The porcine complex or hCore together with 10 hE2:hE3BP samples (0.1 to 1 meq of hE3BP mixed with 1.2 μM hE2) were loaded onto three precast gels ("Any kD Mini-PROTEAN TGX Stain-Free Protein Gels", Bio-Rad) and run simultaneously for 2.5 (pPDHc) or

1.5 hours (hCore) at 180 V. Protein bands were visualized by Coomassie staining (Instant Blue, Sigma-Aldrich) or fluorescence of stain-free gels upon exposure to UV light (Gel Logic 212 PRO, Carestream). Band intensities were determined with GelAnalyzer 19.1 (55) and calibration curves for the E2/E3BP ratio were prepared with OriginPro 2018 (OriginLab).

Analytical ultracentrifugation

Sedimentation experiments were performed at 20°C in an Optima XL-I analytical ultracentrifuge (Beckman Coulter). For sedimentation velocity analytical ultracentrifugation, samples containing purified hE3BP at different initial concentrations (400 μl) or reference buffers (420 μl) were equilibrated at 20°C in the resting An-60 Ti rotor for at least 2 hours before acceleration to 60,000 rpm. Analytical cells were assembled using double-sector 12-mm aluminum centerpieces and quartz windows. Radial scans of the cells were obtained by measuring absorbance at 230 nm, with a single acquisition in continuous mode and a step size of 0.003 cm. Data were analyzed in SEDFIT according to a continuous sedimentation coefficient distribution model [c(s)] (56), with frictional ratio, meniscus, and baseline as fitting parameters. Calculated sedimentation coefficient distributions were normalized using GUSSE (57).

For sedimentation equilibrium experiments, 140 μl of samples containing purified hE3BP at different initial concentrations were successively centrifuged at 8000, 19,000, and 30,000 rpm in an An-50 Ti rotor using analytical cells containing double-sector 12-mm charcoal-filled Epon centerpieces and quartz windows. For each rotor speed, radial scans were collected every 3 hours by measuring absorbance at 230 nm, with a single acquisition in step mode and a step size of 0.001 cm. Attainment of equilibrium was assessed using the respective option in SEDFIT (58). Datasets were globally fitted using SEDPHAT (59) assuming the presence of three noninteracting species. Confidence limits for the fitted molecular masses were determined using F statistics as implemented in SEDPHAT ($P = 0.95$).

Values for buffer density, buffer viscosity, and the partial specific volume of hE3BP (0.7455 ml/g) were calculated using SEDNTERP (60). Protein samples were dissolved in 20 mM Hepes-NaOH, 140 mM KCl, 10 mM NaCl, and 1 mM MgCl_2 (pH 7.4).

Binding stoichiometry of peripheral subunits

Samples for titration experiments were prepared by mixing a constant amount of hCore or the hE2 60-mer with variable concentrations of hE1 or hE3 in PB buffer (total volume: 35 μl). Aliquots of 25 μl were then subjected to gel filtration in PB buffer on a Superdex 200 Increase 5/150 GL column (Cytiva) connected to a 1260 Infinity HPLC System (Agilent Technologies). Eluted proteins were detected via absorption at 280 and 450 nm. Peak areas were quantified with OpenLab CDS (Agilent Technologies) and titration profiles were analyzed with OriginPro 2018.

Association kinetics

Kinetics of association of hPDHc components were recorded at 25°C and pH 7.4 in PB buffer with a SX20 stopped-flow instrument (Applied Photophysics) via the increase in light scattering at 320 nm or (in the case of hE3 binding) quenching of FAD fluorescence above 475 nm (excitation at 450 nm), using identical initial concentrations of respective binding sites in the core particle and hE1 heterotetramers or hE3 homodimers. The data were fitted according to second-order kinetics with OriginPro 2018 using Eq. 1,

$$S = \frac{S_{\infty} - (S_{\infty} - S_0)}{[P]_0 \cdot k \cdot t + 1} \quad (1)$$

where S is the recorded signal, S_{∞} is the signal at full occupancy, S_0 is the signal in the absence of hE1 or hE3, $[P]_0$ is the initial concentration of binding sites for the respective peripheral subunit, k is the apparent second-order rate constant, and t is the reaction time.

Enzymatic activity assay

Enzymatic activity of hPDHc, pPDHc, pOGDHc, and pBCKDHc was measured at 25°C via the increase in NADH absorption at 340 nm. The reactions [20 mM Hepes-NaOH (pH 7.4), 140 mM KCl, 10 mM NaCl, 1 mM MgCl₂, 5 mM α -keto acid, 0.2 mM ThDP, 2.6 mM DTT, 0.2 mM CoA, and 2.5 mM NAD⁺] were initiated either by addition of enzyme or the respective keto acid. Initial velocities were recorded on a Cary 300 spectrophotometer (Agilent Technologies) or on a Synergy 2 plate reader (Agilent Technologies) using autoinjection. After plotting initial velocities against the concentration of peripheral subunit added to hCore (2.0 nM), binding affinities of hE1/hE3 to the hCore were determined according to a noncooperative binding equilibrium model in OriginPro 2018 using Eq. 2

$$v_0 = v_{0\infty} \cdot \frac{[P]_0 + [L]_0 + K_d - \sqrt{([P]_0 + [L]_0 + K_d)^2 - 4 \cdot [P]_0 \cdot [L]_0}}{2 \cdot [P]_0} \quad (2)$$

where v_0 is the recorded initial velocity, $v_{0\infty}$ is the initial velocity at saturation with hE1 or hE3, $[P]_0$ is the total concentration of binding sites for hE1 or hE3 within hCore, $[L]_0$ is the total concentration of hE1 or hE3, and K_d is the dissociation constant.

NS-EM sample preparation and data collection

Quantifoil copper EM grids (300 mesh) coated with a carbon film were used for specimen preparation as previously described (61). Briefly, the sample (4 μ l) was applied to glow-discharged (PELCO easiGlow, Ted Pella, negative, 25 mA, 30 s) grids, followed by 1 min of incubation, blotting with a filter paper, washing in two droplets of water, and sequential staining in two droplets of 2% (w/v) aqueous uranyl acetate. After 1- to 2-min incubation, the grid was blotted and air-dried. NS grids were screened with Morgagni 268 (100 kV) and imaged with Tecnai F20 (200 kV, equipped with FEG) transmission electron microscopes (Thermo Fisher Scientific). For automated data collection, the Tecnai F20 microscope, equipped with the Falcon II direct electron detector, was used. Images were recorded with the EPU software (Thermo Fisher Scientific) as single micrographs at a dose of approximately 50 e⁻/Å² with a resulting pixel size of 1.32 Å/px at 80,000-fold magnification. The targeted defocus was set in the range of -2.5 to -1.0 μ m with 0.3- μ m increments.

NS-EM single-particle analysis

NS-EM datasets were processed in cryoSPARC v3.2 (62) and Relion 3.1 (63, 64) software. For each dataset, the CTF was estimated using the Patch CTF Estimation job in cryoSPARC. Then, 100 to 200 particles were manually selected, extracted, and 2D-classified to obtain 2D class averages, which were later used as references for automated particle picking from all the micrographs using the cryoSPARC Template Picker. The selected particles were extracted

with 400-px box size and binned to 5.28 Å/px, imported (65) to Relion, and subjected to several rounds of 2D classification to remove false-positive picks.

Cryo-EM sample preparation and data collection

hE2 and hCore particles were prepared with the following procedure. Freshly purified complexes (400 nM in 200 μ l PB buffer) were cross-linked during 10 to 40% (w/v) sucrose gradient ultracentrifugation [SW 55 (Beckman Coulter), 30,000 rpm, 16 hours, 4°C] with a 0 to 0.15% glutaraldehyde gradient [GraFix (36, 37)]. Gradients were manually fractionated and tested for absorption at 220 and 280 nm (NanoDrop One spectrometer, Thermo Fisher Scientific). Covalent cross-linking was confirmed by SDS-PAGE. The quality and approximate concentration of particles on the grid were assessed by NS-EM. The sample was plunge-frozen on Quantifoil R2/2 holey carbon copper grids covered with an in-house prepared thin carbon layer (~1.0 to 1.5 nm) using a Vitrobot Mark IV (Thermo Fisher Scientific). The “floating carbon” technique (38, 39) was used for freezing: A thin carbon layer was floated on the sample for 1 min, recovered with the grid, and rapidly mounted in the Vitrobot. PB buffer (5 μ l) was then applied on the carbon side, immediately blotted for 1 to 4 s, and plunge-frozen in liquid ethane.

The cryo-EM data were collected on two Titan Krios TEM microscopes (Thermo Fisher Scientific), operated at 300 kV, and equipped with K2 Summit (hE2 dataset) or K3 (hCore dataset) direct electron detectors (Gatan), both operating in counting mode, and using a slit width of 20 eV on a GIF-Quantum energy filter (Gatan). The automated data collection was conducted with the EPU software (Thermo Fisher Scientific). The hE2 dataset of 4128 movies was recorded with a final pixel size of 0.84 Å/px and a total dose of 80 e⁻/Å². The same dose was used for the hCore sample, where 27,156 movies were recorded with a pixel size of 0.51 Å/px. For both samples, the targeted defocus was set between -1.6 and -2.8 μ m with 0.2- μ m increments.

Cryo-EM single-particle analysis

Cryo-EM datasets of hE2 and hCore were processed in cryoSPARC v3.2 and Relion 3.1 software with identical early processing steps. First, motion-corrected and dose-weighted [MotionCor2 (66)] movies were imported into cryoSPARC where the Patch CTF Estimation was run to estimate the CTF, astigmatism, and relative ice thickness. Following the manual removal of poor-quality micrographs, 100 to 200 particles were manually picked and subjected to 2D classification. The best 2D class averages were used for the automated particle picking on 100 randomly selected micrographs using the template picker. After particle extraction and 2D classification procedures, about 2000 particles corresponding to the best 2D class averages (representing various particle orientations) were used to train the Topaz model (67, 68). Particles were then picked from the total set of micrographs using the trained model, extracted, and subjected to multiple rounds of 2D classification to remove false-positive picks. The particles that corresponded to the selected 2D class averages were imported (65) into Relion where 3D reconstructions were performed in tetrahedral symmetry unless stated otherwise.

For the hE2 dataset, ~120,000 particles (300-px box size; 1.29 Å/px) were selected for further processing. After ab initio model creation in Relion (one class; default parameters), the 3D refinement (7.5° initial global searches) was performed yielding a 3.4-Å map (a

Relion-generated mask covering the full 60-mer was used). The particle stack was then subjected to 3D classification with 1.8° sampling over a 5° range, and the regularization parameter $T = 25$, to separate classes of different diameters (breathing states). The 3D class average of the highest quality (~80,000 particles) was selected and subjected to 3D refinement (7.5° initial global searches), generating a final map at 3.3-Å resolution. The map was filtered on the basis of local resolution estimates in cryoSPARC. To compare 3D reconstructions solved in tetrahedral and icosahedral symmetries, the hE2 dataset was additionally processed with an imposed icosahedral symmetry using an analogous pipeline.

For the hCore dataset, ~513,000 particles (350-px box size; 1.09 Å/px) were selected for further processing. Following the same pipeline as described for the hE2 dataset, the *ab initio* model and consensus 3D refinement were performed in Relion, yielding a map at 3.4-Å resolution. The particle stack was subjected to 3D classification with one class, global angular searches (sampling interval of 3.7° and finer), and the regularization parameter $T = 50$. The obtained 3D class average was then used as a reference in the 3D refinement with local searches of 1.8 Å (map at 3.4-Å resolution). Next, the aligned particles were re-extracted with the binned pixel size (250-px box size; 1.53 Å/px). The new particle stack was refined again and subjected to signal subtraction, using a mask created in UCSF Chimera (69) by map segmentation (70), and symmetrized in Relion (*relion_image_handler*), covering the putative location of six hE3BP pairs (fig. S7A). Following signal subtraction, 3D classification without alignments with the same symmetrized mask, four classes, and $T = 25$ was performed. The particles corresponding to the 3D class average that displayed features characteristic of hE3BP were referred back to their “non-subtracted signal” equivalents and locally refined (mask over a full 60-mer; 3.7° local searches). The procedure of signal subtraction, 3D classification, and local 3D refinement was repeated two more times resulting in the final pool of ~144,000 particles. The refined map and particles were imported into cryoSPARC where the local refinement (20° local searches) to 3.7 Å and filtering based on local resolution estimates were done.

Model building and refinement

The structure of the homo-60-mer of the hE2 catalytic domain [Protein Data Bank (PDB) ID: 6CT0 (23)] was rigid-body-fitted to the tetrahedral cryo-EM map of the hE2 60-mer using UCSF Chimera. After an (AU) had been identified in our map using Phenix (*phenix.map_box*) (71), the five hE2 chains of the model corresponding to the AU (PDB ID: 6CT0) were extracted in PyMOL (The PyMOL Molecular Graphics System, Version 2.4 Schrödinger, LLC). For the hE2 structure, the homopentamer was subjected to five cycles of real-space refinement in Phenix (*phenix.real_space_refine*) (72) with standard parameters, using protein secondary structure and side chain rotamer restraints. After manual real-space refinement in Coot (73) to reduce the clashscore and rotamer outliers, the model was once again real-space-refined in Phenix with the same parameters. The full 60-mer was then reconstructed by applying tetrahedral symmetry in Phenix (*phenix.apply_ncs*) and subjected to five cycles of real-space refinement (Phenix), where noncrystallographic symmetry (NCS), secondary structure, and side chain rotamer restraints were used.

For the hCore model, the homopentamer of hE2 refined for the hE2 60-mer structure was used. One out of five hE2 chains was then

substituted with an AlphaFold-predicted model (74, 75) of hE3BP, generating five potential heteropentamers. Each of the models was rigid-body-fitted to the tetrahedral cryo-EM map of hCore using UCSF Chimera, and then refined in real space in Phenix with the same parameters used for hE2. They were further evaluated in Coot by correlation with the map. Model 3 (Fig. 5D) was selected as representing the closest fit and subsequently modified in Coot by introducing minor changes to reduce the clashscore and rotamer outliers. This heteropentamer was then refined in Phenix and symmetrized (as described above), followed by the final real-space refinement imposing NCS, secondary structure, and side chain rotamer restraints. Models were validated using the comprehensive validation tool in Phenix (*phenix.validation_cryoem*) wherein MolProbity (76) generated model statistics. All figures of models and cryo-EM maps were prepared with UCSF ChimeraX (77, 78).

Native mass spectrometry

For all MS experiments, Milli-Q water (Millipore, Bedford) was used. All samples were sprayed from 200 mM ammonium acetate (pH 8.0, adjusted with NH₄OH). Proteins were freshly prepared and purified before the analysis to prevent sample degradation.

Approximately 5 µl of each solution was sprayed from borosilicate capillaries of ~1-µm inside diameter (B100-75-10, Sutter Instruments), prepared in-house using a micropipette puller (P-1000, Sutter Instruments), and fitted with a platinum wire. Mass spectra were acquired in positive mode on a “SELECT SERIES Cyclic IMS” mass spectrometer (Waters Corporation). This device is equipped with a 32,000 m/z quadrupole filter and an electron capture dissociation cell located in the transfer (post-IMS) region. The mass range was set to m/z 1000 to 32,000 with a 1-s scanning rate. A 50:50 acetonitrile:water solution of a 20 µM cesium iodide (99.999%, Fluka) was used as a calibration solution for the entire mass range. The essential parameters of the mass spectrometer excluding IMS parts operating in “V-mode” were as follows: capillary voltage, 0.8 to 1.6 kV; sampling cone, 20 V; source offset, 30 V; source temperature, 28°C; trap collision energy, 5 V; transfer collision energy, 5 V. The settings required to perform native MS experiments on cIM-MS instruments have been detailed previously (79). All data were acquired and processed using MassLynx (v4.2, Waters).

Cross-linking mass spectrometry—Sample processing, analysis, and identification of cross-linked peptides by database search

The hCore complexes (300 nM) were cross-linked in PB buffer by incubation with 0.5 mM BS³, (Thermo Fisher Scientific) for 30 min at room temperature. The reaction was quenched by the addition of 50 mM tris-HCl (pH 8.0) and incubation for 15 min at room temperature. The cross-linking of the subunits was confirmed via SDS-PAGE and the quality of the resulting particles was verified using NS-EM.

Analysis of cross-links followed a previously published protocol (80). The cross-linked and quenched sample was evaporated to dryness in a vacuum centrifuge and the residue was dissolved in 8 M urea (50 µl). A 50 mM stock solution of tris(2-carboxyethyl)phosphine (2.5 µl) was added and the sample was incubated for 30 min at 37°C. After cooling to room temperature, 2.5 µl of a 100 mM iodoacetamide solution was added and the sample was left for 30 min at room temperature in the dark. Twenty-five microliters of a 150 mM ammonium bicarbonate solution was added to reduce the urea concentration to 5.5 M, and 500 ng of endoproteinase Lys-C (Wako)

was added, resulting in an enzyme-to-substrate ratio of approximately 1:100. Lys-C digestion was allowed to proceed for 2.5 hours at 37°C, this was followed by another dilution step to reduce the urea concentration to 1 M (addition of 320 μ l 50 mM ammonium bicarbonate) and addition of 1 μ g of trypsin (Promega), corresponding to an enzyme-to-substrate ratio of approximately 1:50. Trypsin digestion was performed at 37°C overnight; the sample was then acidified by adding formic acid (2% final concentration) and purified by solid-phase extraction using a SepPak tC18 cartridge (50 mg, Waters).

The purified sample was fractionated by peptide SEC on an ÄKTA micro system equipped with a Superdex 30 Increase column (300 $\times$ 3.2 mm, Cytiva) and a mobile phase of water/acetonitrile/trifluoroacetic acid (70:30:0.1, v/v/v). Four 100 μ l fractions corresponding to an elution volume of 0.9 to 1.3 ml were collected for mass spectrometric analysis and dried.

Two injections of 20% of each fraction were performed for liquid chromatography–tandem MS on a system consisting of an Easy nLC-1200 HPLC and an Orbitrap Fusion Lumos mass spectrometer (both Thermo Fisher Scientific). An Acclaim PepMap RSLC column (250 mm $\times$ 75 μ m, Thermo Fisher Scientific) was operated at a flow rate of 300 nl/min with mobile phases A = water/acetonitrile/formic acid (98:2:0.15, v/v/v) and B = acetonitrile/water/formic acid (98:2:0.15, v/v/v). The gradient was set to 11 to 40% B in 60 min.

The mass spectrometer was operated in data-dependent acquisition mode with a cycle time of 3 s (top speed mode). Precursor ion scans were performed in the Orbitrap analyzer at 120,000 resolution. The most abundant precursors with charge states between +3 and +7 were fragmented by CID in the linear ion trap at a collision energy of 35%. Fragment ions were detected in the Orbitrap at 30,000 resolution.

MS/MS data acquired in the proprietary .raw format were converted into .mzXML format using msconvert, part of the ProteoWizard package (81). The spectra were analyzed using xQuest, version 2.1.5, available from https://gitlab.ethz.ch/leitner_lab using the original scoring scheme from Rinner *et al.* (82) and Walzthoeni *et al.* (83). Searches were performed against a database containing the sequences of the two target proteins, hE2 and hE3BP, and seven contaminant proteins from *E. coli* and *Homo sapiens* (keratins) were identified in a preliminary search. The reversed sequences of the proteins were used as a decoy database to determine the false discovery rate (FDR). Search parameters included the following: enzyme = trypsin, maximum number of missed cleavages = 2, peptide length = 4 to 40 residues, fixed modification = carbamidomethylation on Cys, variable modification = oxidation on Met, cross-link mass shift = 138.06808 Da, monolink/dead-end mass shifts = 156.07864 and 155.09643 Da, cross-linking sites = Lys and N terminus, MS mass error tolerance = ± 15 ppm, and MS/MS error tolerance = ± 20 ppm. Top-scoring spectra were manually assessed and only considered further if they fulfilled the following criteria: fraction of total ion current assigned ≥ 0.15 , and minimum number of bond cleavages per peptide = 4 or 3 consecutive ones. The FDR for the validated candidates was adjusted to $\leq 5\%$ at the nonredundant peptide pair level with the help of the decoy database hits.

Supplementary Materials

This PDF file includes:

Figs. S1 to S12

Table S1

Legend for movie S1

Legend for table S2

Uncropped gels

Other Supplementary Material for this manuscript includes the following:

Movie S1

Table S2

REFERENCES AND NOTES

1. M. S. Patel, T. E. Roche, R. A. Harris, *Alpha-Keto Acid Dehydrogenase Complexes* (Birkhäuser, ed. 1, 1996).
2. L. J. Reed, Multienzyme complexes. *Acc. Chem. Res.* **7**, 40–46 (1974).
3. L. J. Reed, A trail of research from lipoic acid to α -keto acid dehydrogenase complexes. *J. Biol. Chem.* **276**, 38329–38336 (2001).
4. R. N. Perham, Domains, motifs, and linkers in 2-oxo acid dehydrogenase multienzyme complexes: A paradigm in the design of a multifunctional protein. *Biochemistry* **30**, 8501–8512 (1991).
5. M. S. Patel, T. E. Roche, Molecular biology and biochemistry of pyruvate dehydrogenase complexes. *FASEB J.* **4**, 3224–3233 (1990).
6. A. Imbard, A. Boutron, C. Vequaud, M. Zater, P. de Lonlay, H. Ogier de Baulny, C. Barnerias, M. Miné, C. Marsac, J. M. Saudubray, M. Brivet, Molecular characterization of 82 patients with pyruvate dehydrogenase complex deficiency. Structural implications of novel amino acid substitutions in E1 protein. *Mol. Genet. Metab.* **104**, 507–516 (2011).
7. K. P. Patel, T. W. O'Brien, S. H. Subramony, J. Shuster, P. W. Stacpoole, The spectrum of pyruvate dehydrogenase complex deficiency: Clinical, biochemical and genetic features in 371 patients. *Mol. Genet. Metab.* **105**, 34–43 (2012).
8. L. R. Gray, S. C. Tompkins, E. B. Taylor, Regulation of pyruvate metabolism and human disease. *Cell. Mol. Life Sci.* **71**, 2577–2604 (2014).
9. C. L. Quinlan, I. V. Perevoschikova, R. L. S. Goncalves, M. Hey-Mogensen, M. D. Brand, The determination and analysis of site-specific rates of mitochondrial reactive oxygen species production. *Methods Enzymol.* **526**, 189–217 (2013).
10. C. L. Quinlan, R. L. S. Goncalves, M. Hey-Mogensen, N. Yadava, V. I. Bunik, M. D. Brand, The 2-oxoacid dehydrogenase complexes in mitochondria can produce superoxide/hydrogen peroxide at much higher rates than complex I. *J. Biol. Chem.* **289**, 8312–8325 (2014).
11. L. J. Reed, R. M. Oliver, Structure-function relationships in pyruvate and α -ketoglutarate dehydrogenase complexes. *Adv. Exp. Med. Biol.* **148**, 231–241 (1982).
12. T. Wagenknecht, R. Grassucci, D. Schaak, Cryoelectron microscopy of frozen-hydrated α -ketoacid dehydrogenase complexes from *Escherichia coli*. *J. Biol. Chem.* **265**, 22402–22408 (1990).
13. R. A. Harris, M. M. Bowker-Kinley, B. Huang, P. Wu, Regulation of the activity of the pyruvate dehydrogenase complex. *Adv. Enzyme Regul.* **42**, 249–259 (2002).
14. M. S. Patel, N. S. Nemeria, W. Furey, F. Jordan, The pyruvate dehydrogenase complexes: Structure-based function and regulation. *J. Biol. Chem.* **289**, 16615–16623 (2014).
15. L. J. Reed, M. L. Hackert, Structure-function relationships in dihydrolipoamide acyltransferases. *J. Biol. Chem.* **265**, 8971–8974 (1990).
16. M. Koike, L. J. Reed, W. R. Carroll, Alpha-keto acid dehydrogenation complexes: IV. Resolution and reconstitution of the *Escherichia coli* pyruvate dehydrogenation complex. *J. Biol. Chem.* **238**, 30–39 (1963).
17. A. Mattevi, G. Obmolova, E. Schulze, K. H. Kalk, A. H. Westphal, A. De Kok, W. G. J. Hol, Atomic structure of the cubic core of the pyruvate dehydrogenase multienzyme complex. *Science* **255**, 1544–1550 (1992).
18. K. Chandrasekhar, J. Wang, P. Arjunan, M. Sax, Y. H. Park, N. S. Nemeria, S. Kumaran, J. Song, F. Jordan, W. Furey, Insight to the interaction of the dihydrolipoamide acetyltransferase (E2) core with the peripheral components in the *Escherichia coli* pyruvate dehydrogenase complex via multifaceted structural approaches. *J. Biol. Chem.* **288**, 15402–15417 (2013).
19. S. Meinhold, R. Zdanowicz, C. Giese, R. Glockshuber, Dimerization of a 5-kDa domain defines the architecture of the 5-MDa gammaproteobacterial pyruvate dehydrogenase complex. *Sci. Adv.* **10**, eadj6358 (2024).
20. S. Vijayakrishnan, S. M. Kelly, R. J. C. C. Gilbert, P. Callow, D. Bhella, T. Forsyth, J. G. Lindsay, O. Byron, Solution structure and characterisation of the human pyruvate dehydrogenase complex core assembly. *J. Mol. Biol.* **399**, 71–93 (2010).
21. S. L. Powers-Greenwood, M. Rahmatullah, G. A. Radke, T. E. Roche, Separation of protein X from the dihydrolipoyl transacetylase component of the mammalian pyruvate dehydrogenase complex and function of protein X. *J. Biol. Chem.* **264**, 3655–3657 (1989).
22. M. Rahmatullah, S. Gopalakrishnan, P. C. Andrews, C. L. Chang, G. A. Radke, T. E. Roche, Subunit associations in the mammalian pyruvate dehydrogenase complex. Structure and role of protein X and the pyruvate dehydrogenase component binding domain of the dihydrolipoyl transacetylase component. *J. Biol. Chem.* **264**, 2221–2227 (1989).

23. J. Jiang, F. L. Baiesc, Y. Hiromasa, X. Yu, W. H. Hui, X. Dai, T. E. Roche, Z. H. Zhou, Atomic structure of the E2 inner core of human pyruvate dehydrogenase complex. *Biochemistry* **57**, 2325–2334 (2018).
24. L. G. Korotchkina, M. S. Patel, Mutagenesis studies of the phosphorylation sites of recombinant human pyruvate dehydrogenase. SITE-SPECIFIC REGULATION. *J. Biol. Chem.* **270**, 14297–14304 (1995).
25. E. Kolobova, A. Tuganova, I. Boulatnikov, K. M. Popov, Regulation of pyruvate dehydrogenase activity through phosphorylation at multiple sites. *Biochem. J.* **358**, 69–77 (2001).
26. M. Kato, R. M. Wynn, J. L. Chuang, S. C. Tso, M. Machius, J. Li, D. T. Chuang, structural basis for inactivation of the human pyruvate dehydrogenase complex by phosphorylation: Role of disordered phosphorylation loops. *Structure* **16**, 1849–1859 (2008).
27. S. H. Tareen, M. Kutmon, I. C. Arts, T. M. de Kok, C. T. Evelo, M. E. Adriaens, Logical modelling reveals the PDC-PDK interaction as the regulatory switch driving metabolic flexibility at the cellular level. *Genes Nutr.* **14**, 1–16 (2019).
28. C. A. Brautigam, R. M. Wynn, J. L. Chuang, M. Machius, D. R. Tomchick, D. T. Chuang, Structural insight into interactions between dihydrolipoamide dehydrogenase (E3) and E3 binding protein of human pyruvate dehydrogenase complex. *Structure* **14**, 611–621 (2006).
29. E. M. Ciszak, L. G. Korotchkina, P. M. Dominiak, S. Sidhu, M. S. Patel, Structural basis for flip-flop action of thiamin pyrophosphate-dependent enzymes revealed by human pyruvate dehydrogenase. *J. Biol. Chem.* **278**, 21240–21246 (2003).
30. C. A. Brautigam, R. M. Wynn, J. L. Chuang, D. T. Chuang, Subunit and catalytic component stoichiometries of an in vitro reconstituted human pyruvate dehydrogenase complex. *J. Biol. Chem.* **284**, 13086–13098 (2009).
31. S. Prajapati, D. Haselbach, S. Wittig, M. S. Patel, A. Chari, C. Schmidt, H. Stark, K. Tittmann, Structural and functional analyses of the human PDH complex suggest a “division-of-labor” mechanism by local E1 and E3 clusters. *Structure* **27**, 1124–1136 (2019).
32. Y. Hiromasa, T. Fujisawa, Y. Aso, T. E. Roche, Organization of the cores of the mammalian pyruvate dehydrogenase complex formed by E2 and E2 plus the E3-binding protein and their capacities to bind the E1 and E3 components. *J. Biol. Chem.* **279**, 6921–6933 (2004).
33. S. Hezaveh, A. P. Zeng, U. Jandt, Investigation of core structure and stability of human pyruvate dehydrogenase complex: A coarse-grained approach. *ACS Omega* **2**, 1134–1145 (2017).
34. A. P. Bruins, Mechanistic aspects of electrospray ionization. *J. Chromatogr. A* **794**, 345–357 (1998).
35. C. S. Ho, C. W. K. Lam, M. H. M. Chan, R. C. K. Cheung, L. K. Law, L. C. W. Lit, K. F. Ng, M. W. M. Suen, H. L. Tai, electrospray ionisation mass spectrometry: Principles and clinical applications. *Clin. Biochem. Rev.* **24**, 3–12 (2003).
36. B. Kastner, N. Fischer, M. M. Golas, B. Sander, P. Dube, D. Boehringer, K. Hartmuth, J. Deckert, F. Hauer, E. Wolf, H. Uchtenhagen, H. Urlaub, F. Herzog, J. M. Peters, D. Poerschke, R. Lührmann, H. Stark, GraFix: Sample preparation for single-particle electron cryomicroscopy. *Nat. Methods* **5**, 53–55 (2008).
37. H. Stark, GraFix: Stabilization of fragile macromolecular complexes for single particle Cryo-EM. *Methods Enzymol.* **481**, 109–126 (2010).
38. A. Scialola, F. Mangia, S. Imseng, D. Boehringer, K. Berner, M. Shimobayashi, E. Stüttgen, M. N. Hall, N. Ban, T. Maier, The 3.2-Å resolution structure of human mTORC2. *Sci. Adv.* **6**, eabc1251 (2020).
39. N. de Martín Garrido, W. Fu, K. Ramlal, Z. Zhu, D. Miller, D. Boehringer, C. H. S. Aylett, Direct transfer of electron microscopy samples to wetted carbon and graphene films via a support floatation block. *J. Struct. Biol.* **213**, 107677 (2021).
40. Z. H. Zhou, W. Liao, R. H. Cheng, J. E. Lawson, D. B. McCarthy, L. J. Reed, J. K. Stoops, Direct evidence for the size and conformational variability of the pyruvate dehydrogenase complex revealed by three-dimensional electron microscopy. The “breathing” core and its functional relationship to protein dynamics. *J. Biol. Chem.* **276**, 21704–21713 (2001).
41. Z. H. Zhou, D. B. McCarthy, C. M. O'Connor, L. J. Reed, J. K. Stoops, The remarkable structural and functional organization of the eukaryotic pyruvate dehydrogenase complexes. *Proc. Natl. Acad. Sci. U.S.A.* **98**, 14802–14807 (2001).
42. Y. Kong, D. Ming, Y. Wu, J. K. Stoops, Z. H. Zhou, J. Ma, Conformational flexibility of pyruvate dehydrogenase complexes: A computational analysis by quantized elastic deformational model. *J. Mol. Biol.* **330**, 129–135 (2003).
43. M. Salem, Y. Mauguén, T. Prangé, Revisiting glutaraldehyde cross-linking: The case of the Arg-Lys intermolecular doublet. *Acta Crystallogr. Sect. F Struct. Biol. Cryst. Commun.* **66**, 225–228 (2010).
44. K. Giles, J. Ujma, J. Wildgoose, S. Pringle, K. Richardson, D. Langridge, M. Green, A cyclic ion mobility-mass spectrometry system. *Anal. Chem.* **91**, 8564–8573 (2019).
45. A. Leitner, M. Faini, F. Stengel, R. Aebersold, Crosslinking and mass spectrometry: An integrated technology to understand the structure and function of molecular machines. *Trends Biochem. Sci.* **41**, 20–32 (2016).
46. O. G. L. De Marcucci, G. M. Gibb, J. Dick, J. G. Lindsay, Biosynthesis, import and processing of precursor polypeptides of mammalian mitochondrial pyruvate dehydrogenase complex. *Biochem. J.* **251**, 817–823 (1988).
47. A. Chacinska, C. M. Koehler, D. Milenkovic, T. Lithgow, N. Pfanner, Importing mitochondrial proteins: Machineries and mechanisms. *Cell* **138**, 628–644 (2009).
48. E. Krissinel, K. Henrick, Inference of macromolecular assemblies from crystalline state. *J. Mol. Biol.* **372**, 774–797 (2007).
49. E. Krissinel, Crystal contacts as nature's docking solutions. *J. Comput. Chem.* **31**, 133–143 (2010).
50. C. Wang, C. Ma, S. Chang, H. Wu, C. Yan, J. Chen, Y. Wu, S. An, J. Xu, Q. Han, Y. Jiang, Z. Jiang, H. Gao, X. Zhang, Y. Chang, Architecture of intact mammalian pyruvate dehydrogenase complex revealed by in-situ structural analysis. *bioRxiv* 2024.02.01.578502 (2024). <https://doi.org/10.1101/2024.02.01.578502>.
51. M. Neuenschwander, M. Butz, C. Heintz, P. Kast, D. Hilvert, A simple selection strategy for evolving highly efficient enzymes. *Nat. Biotechnol.* **25**, 1145–1147 (2007).
52. R. K. Scopes, Separation by precipitation, in *Protein Purification* (Springer New York, 1994), pp. 71–101.
53. W. Colón, Analysis of protein structure by solution optical spectroscopy. *Methods Enzymol.* **309**, 605–632 (1999).
54. B. J. Greber, D. Boehringer, A. Leitner, P. Bieri, F. Voigts-Hoffmann, J. P. Erzberger, M. Leibundgut, R. Aebersold, N. Ban, Architecture of the large subunit of the mammalian mitochondrial ribosome. *Nature* **505**, 515–519 (2014).
55. I. Lazar Jr., I. Lazar Sr., GelAnalyzer 19.1, www.gelanalyzer.com.
56. P. Schuck, Size-distribution analysis of macromolecules by sedimentation velocity ultracentrifugation and Lamm equation modeling. *Biophys. J.* **78**, 1606–1619 (2000).
57. C. A. Brautigam, Calculations and publication-quality illustrations for analytical ultracentrifugation data. *Methods Enzymol.* **562**, 109–133 (2015).
58. P. Schuck, H. Zhao, C. A. Brautigam, R. Ghirlando, *Basic Principles of Analytical Ultracentrifugation* (CRC Press, ed. 1, 2015).
59. P. Schuck, On the analysis of protein self-association by sedimentation velocity analytical ultracentrifugation. *Anal. Biochem.* **320**, 104–124 (2003).
60. T. M. Laue, B. Shah, T. M. Ridgeway, S. L. Pelletier, Computer-aided interpretation of sedimentation data for proteins, in *Analytical Ultracentrifugation in Biochemistry and Polymer Science*, S. E. Harding, J. C. Horton, A. J. Rowe, Eds. (Royal Society of Chemistry, 1992), pp. 90–125.
61. M. Ohi, Y. Li, Y. Cheng, T. Walz, Negative staining and image classification—Powerful tools in modern electron microscopy. *Biol. Proced. Online* **6**, 23–34 (2004).
62. A. Punjani, J. L. Rubinstein, D. J. Fleet, M. A. Brubaker, cryoSPARC: Algorithms for rapid unsupervised cryo-EM structure determination. *Nat. Methods* **14**, 290–296 (2017).
63. R. Fernandez-Leiro, S. H. W. Scheres, A pipeline approach to single-particle processing in RELION. *Acta Crystallogr. Sect. D Biol. Crystallogr.* **73**, 496–502 (2017).
64. J. Zivanov, T. Nakane, B. O. Forsberg, D. Kimanius, W. J. H. Hagen, E. Lindahl, S. H. W. Scheres, New tools for automated high-resolution cryo-EM structure determination in RELION-3. *eLife* **7**, e42166 (2018).
65. D. Asarnow, E. Palovcak, Y. Cheng, asarnow/pyem: UCSF pyem v0.5, Zenodo (2019); 10.5281/zenodo.3576630.
66. S. Q. Zheng, E. Palovcak, J.-P. Armache, K. A. Verba, Y. Cheng, D. A. Agard, MotionCor2: Anisotropic correction of beam-induced motion for improved cryo-electron microscopy. *Nat. Methods* **14**, 331–332 (2017).
67. T. Bepler, A. Morin, M. Rapp, J. Brasch, L. Shapiro, A. J. Noble, B. Berger, Positive-unlabeled convolutional neural networks for particle picking in cryo-electron micrographs. *Nat. Methods* **16**, 1153–1160 (2019).
68. T. Bepler, K. Kelley, A. J. Noble, B. Berger, Topaz-Denoise: General deep denoising models for cryoEM and cryoET. *Nat. Commun.* **11**, 1–12 (2020).
69. E. F. Pettersen, T. D. Goddard, C. C. Huang, G. S. Couch, D. M. Greenblatt, E. C. Meng, T. E. Ferrin, UCSF Chimera—A visualization system for exploratory research and analysis. *J. Comput. Chem.* **25**, 1605–1612 (2004).
70. G. D. Pintiile, J. Zhang, T. D. Goddard, W. Chiu, D. C. Gossard, Quantitative analysis of cryo-EM density map segmentation by watershed and scale-space filtering, and fitting of structures by alignment to regions. *J. Struct. Biol.* **170**, 427–438 (2010).
71. D. Liebschner, P. V. Afonine, M. L. Baker, G. Bunkóczi, V. B. Chen, T. I. Croll, B. Hintze, L.-W. Hung, S. Jain, A. J. McCoy, N. W. Moriarty, R. D. Oeffner, B. K. Poon, M. G. Prisant, R. J. Read, J. S. Richardson, D. C. Richardson, M. D. Sammito, O. V. Sobolev, D. H. Stockwell, T. C. Terwilliger, A. G. Urzhumtsev, L. L. Videau, C. J. Williams, P. D. Adams, Macromolecular structure determination using X-rays, neutrons and electrons: Recent developments in Phenix. *Acta Crystallogr. Sect. D Biol. Crystallogr.* **75**, 861–877 (2019).
72. P. V. Afonine, B. K. Poon, R. J. Read, O. V. Sobolev, T. C. Terwilliger, A. Urzhumtsev, P. D. Adams, Real-space refinement in PHENIX for cryo-EM and crystallography. *Acta Crystallogr. Sect. D Biol. Crystallogr.* **74**, 531–544 (2018).

73. P. Emsley, B. Lohkamp, W. G. Scott, K. Cowtan, Features and development of Coot. *Acta Crystallogr. Sect. D Biol. Crystallogr.* **66**, 486–501 (2010).
74. J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Žídek, A. Potapenko, A. Bridgland, C. Meyer, S. A. A. Kohl, A. J. Ballard, A. Cowie, B. Romera-Paredes, S. Nikolov, R. Jain, J. Adler, T. Back, S. Petersen, D. Reiman, E. Clancy, M. Zielinski, M. Steinegger, M. Pacholska, T. Berghammer, S. Bodenstein, D. Silver, O. Vinyals, R. W. Senior, K. Kavukcuoglu, P. Kohli, D. Hassabis, Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
75. M. Varadi, S. Anyango, M. Deshpande, S. Nair, C. Natassia, G. Yordanova, D. Yuan, O. Stroe, G. Wood, A. Laydon, A. Žídek, T. Green, K. Tunyasuvunakool, S. Petersen, J. Jumper, E. Clancy, R. Green, A. Vora, M. Lutfi, M. Figurnov, A. Cowie, N. Hobbs, P. Kohli, G. Kleywegt, E. Birney, D. Hassabis, S. Velankar, AlphaFold Protein Structure Database: Massively expanding the structural coverage of protein-sequence space with high-accuracy models. *Nucleic Acids Res.* **50**, D439–D444 (2022).
76. V. B. Chen, W. B. Arendall, J. J. Headd, D. A. Keedy, R. M. Immormino, G. J. Kapral, L. W. Murray, J. S. Richardson, D. C. Richardson, MolProbity: All-atom structure validation for macromolecular crystallography. *Acta Crystallogr. Sect. D Biol. Crystallogr.* **66**, 12–21 (2010).
77. T. D. Goddard, C. C. Huang, E. C. Meng, E. F. Pettersen, G. S. Couch, J. H. Morris, T. E. Ferrin, UCSF ChimeraX: Meeting modern challenges in visualization and analysis. *Protein Sci.* **27**, 14–25 (2018).
78. E. F. Pettersen, T. D. Goddard, C. C. Huang, E. C. Meng, G. S. Couch, T. I. Croll, J. H. Morris, T. E. Ferrin, UCSF ChimeraX: Structure visualization for researchers, educators, and developers. *Protein Sci.* **30**, 70–82 (2021).
79. J. A. Harrison, A. Pruska, P. Bittner, A. Muck, D. A. Cooper-Shepherd, R. Zenobi, Advancing cyclic ion mobility mass spectrometry methods for studying biomolecules: Toward the conformational dynamics of mega Dalton protein aggregates. *Anal. Chem.* **94**, 12435–12443 (2022).
80. A. Leitner, T. Walzthoeni, R. Aebersold, Lysine-specific chemical cross-linking of protein complexes and identification of cross-linking sites using LC-MS/MS and the xQuest/xProphet software pipeline. *Nat. Protoc.* **9**, 120–137 (2013).
81. M. C. Chambers, B. MacLean, R. Burke, D. Amodei, D. L. Ruderman, S. Neumann, L. Gatto, B. Fischer, B. Pratt, J. Egertson, K. Hoff, D. Kessner, N. Tasman, N. Shulman, B. Frewen, T. A. Baker, M. Y. Brusniak, C. Paulse, D. Creasy, L. Flashner, K. Kani, C. Moulding, S. L. Seymour, L. M. Nuwaysir, B. Lefebvre, F. Kuhlmann, J. Roark, P. Rainer, S. Detlev, T. Hemenway, A. Huhmer, J. Langridge, B. Connolly, T. Chadick, K. Holly, J. Eckels, E. W. Deutsch, R. L. Moritz, J. E. Katz, D. B. Agus, M. MacCoss, D. L. Tabb, P. Mallick, A cross-platform toolkit for mass spectrometry and proteomics. *Nat. Biotechnol.* **30**, 918–920 (2012).
82. O. Rinner, J. Seebacher, T. Walzthoeni, L. Mueller, M. Beck, A. Schmidt, M. Mueller, R. Aebersold, Identification of cross-linked peptides from large sequence databases. *Nat. Methods* **5**, 315–318 (2008).
83. T. Walzthoeni, M. Claassen, A. Leitner, F. Herzog, S. Bohn, F. Förster, M. Beck, R. Aebersold, False discovery rate estimation for cross-linked peptides identified by mass spectrometry. *Nat. Methods* **9**, 901–903 (2012).
84. Y. Perez-Riverol, J. Bai, C. Bandla, D. García-Seisdedos, S. Hewapathirana, S. Kamatchinathan, D. J. Kundu, A. Prakash, A. Frericks-Zipper, M. Eisenacher, M. Walzer, S. Wang, A. Brazma, J. A. Vizcaino, The PRIDE database resources in 2022: A hub for mass spectrometry-based proteomics evidences. *Nucleic Acids Res.* **50**, D543–D552 (2022).

Acknowledgments: We thank P. Bachmann and D. Zyla for IT support, E. Kummer for help with the purification of porcine mitochondria, and J. Stanisich for revision of this manuscript. The cryo-EM data were collected at the Scientific Center for Optical and Electron Microscopy at the ETH Zurich (ScopeM). We thank the ScopeM staff, especially M. Peterrek, for technical support during data collection. We also thank the staff of the Functional Genomics Center Zurich for the intact mass determination of the proteins used in this study, as well as P. Picotti for access to the mass spectrometry instrumentation and laboratory infrastructure. **Funding:** This work was supported by ETH Zurich and the Swiss National Science Foundation (SNSF), grant 310030_201234 to R.G. **Author contributions:** R.Zd. and R.G. started the project and designed experiments. R.Zd. purified the proteins, as well as performed and analyzed all biochemical assays. R.Zd. and P.A. collected cryo-EM data. R.Zd., P.A., and D.B. performed single-particle cryo-EM analysis. A.P. and J.A.H. recorded and analyzed native MS data. C.G. collected and analyzed analytical ultracentrifugation data. A.L. recorded and processed the XL-MS data. R.G. and R.Ze. secured funding for the project. R.Zd. prepared the figures with input from coauthors, and all coauthors contributed to the writing of this manuscript. **Competing interests:** The authors declare that they have no competing interests. **Data and materials availability:** All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials. The structural data reported in this study are available in the Electron Microscopy Data Bank and Protein Data Bank (PDB) under accession codes EMD- 17691 (hE2 I sym), EMD- 17694 (hE2 T sym), EMD- 18616 (hCore map 1), and EMD- 18617 (hCore map 2) and PDB ID 8PIU (hE2 I sym). The cross-linking mass spectrometry proteomics data were deposited to the ProteomeXchange Consortium (<http://proteomecentral.proteomexchange.org>) via the PRIDE partner repository (84) with the dataset identifier PXD045858.

Submitted 11 December 2023

Accepted 5 June 2024

Published 17 July 2024

10.1126/sciadv.adn4582